Supporting Information

Bridging Atomistic Simulations and Neural Dynamics for Predicting the Future of Click-Crosslinked Carboxymethyl Cellulose Hydrogels

Rontu Das, Debashis Kundu *

Department of Chemical Engineering, Institute of Chemical Technology Marathwada Campus, Jalna, Maharashtra, 431203, India

* Corresponding author's email: d.kundu@marj.ictmumbai.edu.in

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Table S1. System Details

Polymer name	Initial box dimension (nm^3)	Final box dimension (nm^3)	Total number of polymer molecules	Total number of water molecules	Total number of molecules
CMC	19.55×19.55×19.55	6.85×6.85×6.85	2166	40000	42166
CMC-N	19.86×19.86×19.86	6.86×6.86×6.86	2278	40000	42278
CMC-NT	19.87×19.87×19.87	6.88×6.88×6.88	2422	40000	42422
CMC-NTC	30.51×30.51×30.51	8.83×8.83×8.83	9400	80000	89400

Table S2. Optimisation of noise_in hyperparameter of reservoir for Hbond data

noise_in	*RMSE* ($\times 10^{-3}$)	R^2
0.00	12.81	0.99
1.00×10^{-8}	12.81	0.99
1.00×10^{-7}	12.81	0.99
1.00×10^{-6}	12.81	0.99
1.00×10^{-5}	12.81	0.99
1.00×10^{-4}	12.81	0.99
1.00×10^{-3}	13.14	0.98
0.01	24.55	0.96
0.10	80.14	0.31
0.20	105.35	-0.61
0.30	121.40	-1.61
0.40	135.49	-2.99
0.50	148.05	-5.00
0.60	158.60	-7.74
0.70	167.01	-11.05
0.80	173.47	-14.6
0.90	178.32	-17.96
1.00	181.94	-20.8
1.50	190.28	-26.22
2.00	192.52	-26.19
3.00	193.18	-26.45
4.00	193.10	-25.78
5.00	193.00	-24.34

Table S3. Optimisation of noise_fb hyperparameter of reservoir for Hbond data

noise_fb	$RMSE(\times 10^{-3})$	R^2
0.00	12.81	0.99
1.00×10^{-5}	12.81	0.99
1.00×10^{-4}	12.81	0.99
1.00×10^{-3}	12.81	0.99
0.01	12.81	0.99
0.10	12.81	0.99
0.20	12.81	0.99
0.30	12.81	0.99
0.40	12.81	0.99
0.50	12.81	0.99
0.60	12.81	0.99
0.70	12.81	0.99
0.80	12.81	0.99
0.90	12.81	0.99
1.00	12.81	0.99
1.50	12.81	0.99
2.00	12.81	0.99
3.00	12.81	0.99
4.00	12.81	0.99
5.00	12.81	0.99

Table S4. Optimisation of noise_rc hyperparameter of reservoir for Hbond data

noise_rc	$RMSE(\times 10^{-3})$	R^2
0.00	12.81	0.99
1.00×10^{-5}	12.80	0.99
1.00×10^{-4}	13.29	0.98
1.00×10^{-3}	17.99	0.98
0.01	32.84	0.92
0.10	102.88	-0.13
0.20	147.35	-3.28
0.30	165.75	-7.14
0.40	173.88	-10.37
0.50	177.94	-12.68
0.60	180.18	-14.23
0.70	181.49	-15.26
0.80	182.32	-15.95
0.90	182.85	-16.42
1.00	183.22	-16.75
1.50	184.03	-17.4
2.00	184.34	-17.46
3	184.67	-17.37
4	184.85	-17.29
5	184.96	-17.24

Table S5. Optimisation of input_bias hyperparameter of reservoir for Hbond data

input_bias	$RMSE(\times 10^{-3})$	R^2
TRUE	12.81	0.99
FALSE	12.37	0.99

Table S6. Optimisation of fb_scaling hyperparameter of reservoir for Hbond data

fb_scaling	$RMSE(\times 10^{-3})$	R^2
1.00×10^{-5}	12.37	0.99
1.00×10^{-4}	12.37	0.99
1.00×10^{-3}	12.37	0.99
0.01	12.37	0.99
0.10	12.37	0.99
0.20	12.37	0.99
0.30	12.37	0.99
0.40	12.37	0.99
0.50	12.37	0.99
0.60	12.37	0.99
0.70	12.37	0.99
0.80	12.37	0.99
0.90	12.37	0.99
1.00	12.37	0.99
1.50	12.37	0.99
2.00	12.37	0.99
3.00	12.37	0.99
4	12.37	0.99
5	12.37	0.99

Table S7. Optimisation of input_connectivity of reservoir for Hbond data

input_connectivity	$RMSE(\times 10^{-3})$	R^2
1.00×10^{-5}	181.87	-1.7×10^{32}
1.00×10^{-4}	181.87	-1.7×10^{32}
1.00×10^{-3}	181.87	-1.7×10^{32}
0.01	181.87	-1.7×10^{32}
0.10	12.37	0.99
0.20	12.36	0.99
0.30	11.34	0.99
0.31	11.58	0.99
0.32	11.58	0.99
0.33	11.90	0.99
0.34	11.90	0.99
0.35	11.90	0.99
0.36	11.21	0.99
0.37	11.21	0.99
0.38	11.21	0.99
0.39	11.29	0.99
0.40	11.29	0.99
0.41	12.22	0.99
0.42	12.22	0.99
0.43	12.22	0.99
0.44	11.67	0.99
0.45	11.67	0.99
0.46	11.67	0.99
0.47	11.71	0.99
0.48	11.71	0.99
0.49	12.07	0.99
0.5	12.07	0.99
0.60	11.53	0.99
0.70	11.55	0.99

0.80	11.79	0.99
0.90	11.59	0.99
1.00	11.12	0.99

Table S8. Optimisation of rc_connectivity of reservoir for Hbond data

rc_connectivity	$RMSE(\times 10^{-3})$	R^2
1.00×10^{-3}	41.39	0.88
0.01	40.21	0.89
0.10	11.21	0.99
0.20	14.42	0.98
0.30	12.99	0.99
0.40	12.43	0.99
0.50	13.15	0.98
0.60	11.75	0.99
0.70	13.31	0.98
0.80	12.22	0.99
0.90	12.04	0.99
1.00	12.40	0.99

Table S9. Optimisation of fb_connectivity of reservoir for Hbond data

fb_connectivity	$RMSE(\times 10^{-3})$	R^2
0.10	11.21	0.99
0.20	11.21	0.99
0.30	11.21	0.99
0.40	11.21	0.99
0.50	11.21	0.99
0.60	11.21	0.99
0.70	11.21	0.99
0.80	11.21	0.99
0.90	11.21	0.99
1.00	11.21	0.99

Table 10. Optimisation of activation function of reservoir for Hbond data

activation function	$RMSE(\times 10^{-3})$	R^2
tanh	11.21	0.99
sigmoid	19.52	0.97
softplus	16.44	0.98
identity	11.16	0.99

Table S11. Optimisation of equation of reservoir for Hbond data

equation	$RMSE(\times 10^{-3})$	R^2
internal	11.21	0.99
external	11.34	0.99

Table S12. Optimisation of ridge hyper parameter for Hbond data

ridge	$RMSE(\times 10^{-3})$	R^2
1.00×10^{-1}	21.56	0.97
1.00×10^{-2}	16.91	0.98
1.00×10^{-3}	14.24	0.98
1.00×10^{-4}	12.55	0.99
1.00×10^{-5}	11.21	0.99
1.00×10^{-6}	10.58	0.99
1.00×10^{-7}	10.71	0.99
1.00×10^{-8}	11.04	0.99
1.00×10^{-9}	11.13	0.99
1.00×10^{-10}	10.64	0.99
1.00×10^{-11}	10.43	0.99
1.00×10^{-12}	10.54	0.99
1.00×10^{-13}	10.57	0.99
1.00×10^{-14}	10.58	0.99
1.00×10^{-15}	21.56	0.99

Table S13. Optimisation of the number of layers in GRU algorithm for Hbond data

number of layers	*RMSE*	R^2
1	0.29	0.02
2	0.29	0.02
3	0.28	0.02
4	0.29	-0.03
5	0.29	-0.04
6	0.29	-0.03
7	0.29	-0.03
8	0.30	-0.07
9	0.30	-0.04
10	0.29	-0.03

Table S14. Optimisation of the hidden size in GRU algorithm for Hbond data

hidden size	*RMSE*	R^2
2	0.29	-0.02
4	0.29	-0.01
8	0.29	0.00
16	0.29	0.02
32	0.29	0.02
64	0.29	0.02
128	0.28	0.02
256	0.29	-0.01
512	0.29	-0.02
1024	0.29	-0.03

Table S15. Optimisation of the learning rate in GRU algorithm for Hbond data

learning rate	*RMSE*	R^2
1.00×10^{-06}	0.30	-0.10
1.00×10^{-05}	0.29	-0.04
1.00×10^{-04}	0.29	0.00
2.00×10^{-04}	0.29	0.01
3.00×10^{-04}	0.29	0.02
4.00×10^{-04}	0.29	0.02
5.00×10^{-04}	0.29	0.02
6.00×10^{-04}	0.29	0.02
7.00×10^{-04}	0.29	0.01
8.00×10^{-04}	0.29	0.02
9.00×10^{-04}	0.29	0.02
1.00×10^{-03}	0.28	0.02
2.00×10^{-03}	0.29	-0.01
3.00×10^{-03}	0.30	-0.04
4.00×10^{-03}	0.29	-0.02
5.00×10^{-03}	0.29	-0.02
6.00×10^{-03}	0.29	-0.01
7.00×10^{-03}	0.28	0.03
8.00×10^{-03}	0.30	-0.09
9.00×10^{-03}	0.32	-0.19
1.00×10^{-02}	0.41	-1.03
1.00×10^{-01}	0.30	-0.07

Table S16. Optimisation of the batch size in GRU algorithm for Hbond data

batch size	*RMSE*	R^2
4	0.43	-1.16
8	0.31	-0.12
16	0.31	-0.15
32	0.28	0.03
64	0.31	-0.17
128	0.27	0.03
256	0.29	0.02
512	0.29	0.00

Table S17. Optimisation of the optimiser in GRU algorithm for Hbond data

optimiser	*RMSE*	R^2
Adam	0.27	0.03
AdamW	0.27	0.04
RMSprop	0.29	-0.03
SGD	0.29	-0.04

Table S18. Optimisation of the data percentage in GRU algorithm for Hbond data

data percentage	*RMSE*	R^2
10	0.32	-0.12
20	0.30	-0.05
30	0.31	-0.09
40	0.29	0.03
50	0.30	-0.01
60	0.24	0.05
70	0.25	0.04
80	0.29	-0.01

Table S19. Optimisation of the number of layers in LSTM algorithm for Hbond data

number of layers	*RMSE*	R^2
1	0.29	0.01
2	0.29	0.02
3	0.30	-0.04
4	0.29	-0.03
5	0.30	-0.04
6	0.30	-0.04
7	0.30	-0.04
8	0.30	-0.04
9	0.30	-0.04
10	0.30	-0.04

Table S20. Optimisation of the hidden size in LSTM algorithm for Hbond data

hidden size	*RMSE*	R^2
2	0.30	-0.04
4	0.29	0.01
8	0.29	-0.02
16	0.29	0.01
32	0.29	0.02
64	0.29	0.02
128	0.29	0.02
256	0.29	0.02
512	0.28	0.02
1024	0.29	-0.01

Table S21. Optimisation of the learning rate in LSTM algorithm for Hbond data

learning rate	*RMSE*	R^2
1.00×10^{-06}	0.30	-0.05
1.00×10^{-05}	0.29	-0.04
1.00×10^{-04}	0.27	0.02
2.00×10^{-04}	0.29	0.01
3.00×10^{-04}	0.29	0.01
4.00×10^{-04}	0.29	0.01
5.00×10^{-04}	0.29	0.00
6.00×10^{-04}	0.29	0.01
7.00×10^{-04}	0.29	0.00
8.00×10^{-04}	0.29	0.00
9.00×10^{-04}	0.29	0.00
1.00×10^{-03}	0.28	0.02
2.00×10^{-03}	0.29	-0.02
3.00×10^{-03}	0.30	-0.04
4.00×10^{-03}	0.29	-0.03
5.00×10^{-03}	0.29	-0.03
6.00×10^{-03}	0.32	-0.24
7.00×10^{-03}	0.29	-0.03
8.00×10^{-03}	0.30	-0.04
9.00×10^{-03}	0.30	-0.04
1.00×10^{-02}	0.37	-0.67
1.00×10^{-01}	0.38	-0.74

Table S22. Optimisation of the batch size in LSTM algorithm for Hbond data

batch size	*RMSE*	R^2
4	0.29	0.00
8	0.27	0.02
16	0.29	0.01
32	0.27	0.02
64	0.29	0.02
128	0.29	0.01
256	0.29	-0.02
512	0.29	-0.03

Table S23. Optimisation of the optimiser in LSTM algorithm for Hbond data

optimiser	*RMSE*	R^2
Adam	0.27	0.02
AdamW	0.29	0.00
RMSprop	0.26	0.03
SGD	0.30	-0.05

Table S24. Optimisation of the data percentage in LSTM algorithm for Hbond data

data percentage	*RMSE*	R^2
10	0.29	-0.12
20	0.28	-0.13
30	0.27	-0.03
40	0.26	0.01
50	0.27	0.04
60	0.25	0.04
70	0.26	0.02
80	0.26	0.01

Table S25. Optimisation of the number of layers in LAG-LLAMA algorithm for Hbond data

number of layers	*RMSE*	R^2
1	0.29	-0.03
2	0.29	-0.03
3	0.29	-0.03
4	0.30	-0.05
5	0.30	-0.04
6	0.30	-0.04
7	0.30	-0.06
8	0.30	-0.04
9	0.30	-0.05
10	0.30	-0.05
11	0.30	-0.05
12	0.29	-0.02
13	0.30	-0.06
14	0.30	-0.07
15	0.31	-0.15
16	0.30	-0.07

Table S26. Optimisation of the d_model in LAG-LLAMA algorithm for Hbond data

d_model	*RMSE*	R^2
4	0.29	-0.03
8	0.28	-0.02
16	0.29	-0.03
32	0.29	-0.03
64	0.29	-0.02
128	0.33	-0.27
256	0.33	-0.26
512	0.30	-0.05
1024	0.32	-0.23

Table S27. Optimisation of the n_heads in LAG-LLAMA algorithm for Hbond data

n_heads	*RMSE*	R^2
1	0.29	-0.03
2	0.30	-0.04
4	0.28	-0.02
8	0.27	-0.01
16	0.28	-0.02

Table S28. Optimisation of the dropout in LAG-LLAMA algorithm for Hbond data

dropout	*RMSE*	R^2
0.01	0.30	-0.06
0.02	0.30	-0.04
0.03	0.30	-0.05
0.04	0.29	-0.03
0.05	0.29	-0.03
0.06	0.30	-0.04
0.07	0.30	-0.04
0.08	0.29	-0.03
0.09	0.29	-0.03
0.1	0.27	-0.01
0.11	0.29	-0.03
0.12	0.30	-0.05
0.13	0.30	-0.05
0.14	0.30	-0.07
0.15	0.30	-0.04
0.16	0.30	-0.05
0.17	0.29	-0.03
0.18	0.30	-0.04
0.19	0.29	-0.03
0.2	0.30	-0.03

Table S29. Optimisation of the learning rate in LAG-LLAMA algorithm for Hbond data

learning rate	*RMSE*	R^2
1.00×10^{-06}	1.12	-14.01
1.00×10^{-05}	0.29	-0.02
1.00×10^{-04}	0.29	0.01
2.00×10^{-04}	0.29	-0.03
3.00×10^{-04}	0.29	0.00
4.00×10^{-04}	0.29	-0.03
5.00×10^{-04}	0.29	0.00
6.00×10^{-04}	0.29	-0.03
7.00×10^{-04}	0.29	-0.03
8.00×10^{-04}	0.29	-0.03
9.00×10^{-04}	0.30	-0.05
1.00×10^{-03}	0.27	-0.01
2.00×10^{-03}	0.29	-0.03
3.00×10^{-03}	0.30	-0.04
4.00×10^{-03}	0.29	-0.03
5.00×10^{-03}	0.26	0.01
6.00×10^{-03}	0.30	-0.06
7.00×10^{-03}	0.30	-0.06
8.00×10^{-03}	0.30	-0.04
9.00×10^{-03}	0.29	-0.03
1.00×10^{-02}	0.30	-0.04
1.00×10^{-01}	0.32	-0.20

Table S30. Optimisation of the batch size in LAG-LLAMA algorithm for Hbond data

batch size	*RMSE*	R^2
4	0.26	0.02
8	0.30	-0.07
16	0.29	-0.03
32	0.26	0.01
64	0.29	-0.03
128	0.29	-0.03
256	0.29	-0.03
512	0.29	-0.03

Table S31. Optimisation of the optimiser in LAG-LLAMA algorithm for Hbond data

optimiser	*RMSE*	R^2
Adam	0.26	0.02
AdamW	0.30	-0.05
RMSprop	0.32	-0.20
SGD	0.31	-0.15

Table S32. Optimisation of the data percentage in LAG-LLAMA algorithm for Hbond data

data percentage	*RMSE*	R^2
10	0.35	-0.40
20	0.32	-0.25
30	0.31	-0.10
40	0.31	-0.15
50	0.30	-0.01
60	0.29	-0.02
70	0.26	0.02
80	0.30	-0.10

Table S33. Optimisation of noise_in hyper parameter for MSD data

noise_in	$RMSE(\times 10^{-3})$	R^2
0	1.49	0.99
1.00×10^{-1}	35.66	0.96
1.00×10^{-2}	13.19	0.99
1.00×10^{-3}	1.63	0.99
1.00×10^{-4}	1.49	0.99
1.00×10^{-5}	1.49	0.99
2.00×10^{-5}	1.49	0.99
3.00×10^{-5}	1.49	0.99
4.00×10^{-5}	1.49	0.99
5.00×10^{-5}	1.49	0.99
6.00×10^{-5}	1.49	0.99
7.00×10^{-5}	1.49	0.99
8.00×10^{-5}	1.49	0.99
9.00×10^{-5}	1.49	0.99
1.00×10^{-6}	1.49	0.99
0.2	40.68	0.94
0.3	48.15	0.92
0.4	59.93	0.88
0.5	76.49	0.82
0.6	97.53	0.71
0.7	122.45	0.56
0.8	150.57	0.34
0.9	181.25	0.07
1	213.86	-0.27
1.5	387.73	-3.04
2	548.21	-7.39
3	775.05	-19.41
4	901.98	-33.8
5	974.03	-48.8

Table S34. Optimisation of noise_fb hyper parameter for MSD data

Noise_fb	$RMSE(\times 10^{-3})$	R^2
0	1.49	0.99
1.00×10^{-1}	1.49	0.99
1.00×10^{-2}	1.49	0.99
1.00×10^{-3}	1.49	0.99
1.00×10^{-4}	1.49	0.99
1.00×10^{-5}	1.49	0.99
0.2	1.49	0.99
0.3	1.49	0.99
0.4	1.49	0.99
0.5	1.49	0.99
0.6	1.49	0.99
0.7	1.49	0.99
0.8	1.49	0.99
0.9	1.49	0.99
1	1.49	0.99
1.5	1.49	0.99
2	1.49	0.99
3	1.49	0.99
4	1.49	0.99
5	1.49	0.99

Table S35. Optimisation of noise_rc hyper parameter MSD data

noise_rc	$RMSE(\times 10^{-3})$	R^2
0	1.49	0.99
1.00×10^{-1}	1036.70	-72.38
1.00×10^{-2}	107.99	0.67
1.00×10^{-3}	18.77	0.98
1.00×10^{-4}	4.98	0.99
1.00×10^{-5}	1.50	0.99
1.00×10^{-6}	1.49	0.99
1.00×10^{-7}	1.49	0.99
1.00×10^{-8}	1.49	0.99
0.2	1119.53	-153.02
0.3	1136.03	-188.57
0.4	1141.98	-205.04
0.5	1144.87	-214.23
0.6	1146.54	-220.15
0.7	1147.61	-224.21
0.8	1148.35	-227.06
0.9	1148.90	-229.15
1	1149.31	-230.76
1.5	1150.43	-235.49
2	1150.91	-237.9
3	1151.33	-240.11
4	1151.50	-241.05
5	1151.59	-241.56

Table S36. Optimisation of noise_type hyper parameter MSD data

noise_type	$RMSE(\times 10^{-3})$	R^2
normal	1.49	0.99
uniform	1.49	0.99

Table S37. Optimisation of input_bias hyper parameter MSD data

noise_type	$RMSE(\times 10^{-3})$	R^2
True,	1.49	0.99
False	1.49	0.99

Table S38. Optimisation of bias_scaling hyper parameter MSD data

bias_scaling	$RMSE(\times 10^{-3})$	R^2
1.00×10^{-1}	1.49	0.99
1.00×10^{-2}	1.49	0.99
1.00×10^{-3}	1.49	0.99
1.00×10^{-4}	1.49	0.99
1.00×10^{-5}	1.49	0.99
2.00×10^{-1}	1.49	0.99
3.00×10^{-1}	1.49	0.99
4.00×10^{-1}	1.49	0.99
5.00×10^{-1}	1.49	0.99
6.00×10^{-1}	1.49	0.99
7.00×10^{-1}	1.49	0.99
8.00×10^{-1}	1.49	0.99
9.00×10^{-1}	1.49	0.99
1.00	1.49	0.99
1.50	1.49	0.99
2	1.49	0.99
3	1.49	0.99
4	1.49	0.99
5	1.49	0.99

Table S39. Optimisation of fb_scaling hyper parameter MSD data

fb_scaling	$RMSE(\times 10^{-3})$	R^2
1.00×10^{-1}	1.49	0.99
1.00×10^{-2}	1.49	0.99
1.00×10^{-3}	1.49	0.99
1.00×10^{-4}	1.49	0.99
1.00×10^{-5}	1.49	0.99
2.00×10^{-1}	1.49	0.99
3.00×10^{-1}	1.49	0.99
4.00×10^{-1}	1.49	0.99
5.00×10^{-1}	1.49	0.99
6.00×10^{-1}	1.49	0.99
7.00×10^{-1}	1.49	0.99
8.00×10^{-1}	1.49	0.99
9.00×10^{-1}	1.49	0.99
1.00	1.49	0.99
1.50	1.49	0.99
2	1.49	0.99
3	1.49	0.99
4	1.49	0.99
5	1.49	0.99

Table S40. Optimisation of input_connectivity hyper parameter MSD data

input_connectivity	$RMSE(\times 10^{-3})$	R^2
1.00×10^{-1}	1.49	0.99
1.00×10^{-2}	1150.91	-9.5×10^{8}
1.00×10^{-3}	1150.91	-9.5×10^{8}
1.00×10^{-4}	1150.91	-9.5×10^{8}
1.00×10^{-5}	1150.91	-9.5×10^{8}
0.11	1.49	0.99
0.12	1.49	0.99
0.13	1.49	0.99
0.14	1.49	0.99
0.15	1.49	0.99
0.16	1.49	0.99
0.17	1.49	0.99
0.18	1.49	0.99
0.19	1.49	0.99
0.20	1.49	0.99
0.30	53.37	0.88
0.40	53.37	0.88
0.50	5.37	0.99
0.60	5.84	0.99
0.70	5.84	0.99
0.80	5.92	0.99
0.90	5.92	0.99
1.00	38.95	0.94

Table S41. Optimisation of the rc_connectivity hyper parameter MSD data

rc_connectivity	$RMSE(\times 10^{-3})$	R^2
0.10	1.49	0.99
0.11	1.49	0.99
0.12	1.49	0.99
0.20	9.59	0.99
0.30	77.99	0.72
0.40	9.89	0.99
0.50	11.15	0.99
0.60	3.68	0.99
0.70	133.38	0.01
0.80	1.79	0.99
0.90	49.91	0.9
1.00	41.94	0.94

Table S42. Optimisation of the fb_connectivity hyper parameter MSD data

fb_connectivity	$RMSE(\times 10^{-3})$	R^2
0.10	1.49	0.99
0.20	1.49	0.99
0.30	1.49	0.99
0.40	1.49	0.99
0.50	1.49	0.99
0.60	1.49	0.99
0.70	1.49	0.99
0.80	1.49	0.99
0.90	1.49	0.99
1.00	1.49	0.99

Table S43. Optimisation of the activation function for MSD data

activation function	$RMSE(\times 10^{-3})$	R^2
tanh	1.49	0.99
sigmoid	7.39	0.99
softplus	54.47	0.91
identity	5.40	0.99

Table S44. Optimisation of the equation in reservoir for MSD data

equation	$RMSE(\times 10^{-3})$	R^2
internal	1.49	0.99
external	10.07	0.99

Table S45. Optimisation of the ridge hyperparameter for MSD data

ridge	$RMSE(\times 10^{-3})$	R^2
1.00×10^{-1}	19.32	0.98
1.00×10^{-2}	27.13	0.97
1.00×10^{-3}	22.75	0.98
1.00×10^{-4}	7.55	0.99
1.00×10^{-5}	1.49	0.99
1.00×10^{-6}	1.59	0.99
1.00×10^{-7}	1.58	0.99
1.00×10^{-8}	1.59	0.99
1.00×10^{-9}	1.60	0.99
1.00×10^{-10}	1.61	0.99
1.00×10^{-11}	1.61	0.99
1.00×10^{-12}	1.61	0.99
1.00×10^{-13}	1.61	0.99
1.00×10^{-14}	1.61	0.99
1.00×10^{-15}	1.61	0.99

Table S46. Optimisation of the number of layers in GRU algorithm for MSD data

number of layers	*RMSE*	R^2
1	0.12	0.08
2	0.12	-3.70
3	0.11	0.57
4	0.44	-3.13
5	0.36	-1.23
6	0.38	-3.10
7	0.39	-3.72
8	0.39	-4.57
9	0.36	-1.29
10	0.38	-3.40

Table S47. Optimisation of the hidden size in GRU algorithm for MSD data

hidden size	*RMSE*	R^2
2	0.42	-206.28
4	0.12	-15.96
8	0.08	-4.52
16	0.07	-0.74
32	0.12	-3.07
64	0.06	0.92
128	0.11	0.57
256	0.13	-0.17
512	0.07	0.29
1024	0.03	0.81

Table S48. Optimisation of the learning rate in GRU algorithm for MSD data

learning rate	*RMSE*	R^2
1.00×10^{-06}	0.85	-753.30
1.00×10^{-05}	0.69	-476.88
1.00×10^{-04}	0.02	0.72
2.00×10^{-04}	0.04	0.90
3.00×10^{-04}	0.04	0.76
4.00×10^{-04}	0.10	-1.53
5.00×10^{-04}	0.11	-0.38
6.00×10^{-04}	0.07	-0.32
7.00×10^{-04}	0.08	-0.76
8.00×10^{-04}	0.04	0.95
9.00×10^{-04}	0.15	-16.33
1.00×10^{-03}	0.06	0.92
2.00×10^{-03}	0.11	-1.33
3.00×10^{-03}	0.53	-37.14
4.00×10^{-03}	0.34	-5.99
5.00×10^{-03}	0.32	-2.77
6.00×10^{-03}	0.34	-4.96
7.00×10^{-03}	0.34	-0.75
8.00×10^{-03}	0.35	-5.63
9.00×10^{-03}	0.35	-1.08
1.00×10^{-02}	0.44	-12.60
1.00×10^{-01}	0.53	-36.11

Table S49. Optimisation of the batch size in GRU algorithm for MSD data

batch size	*RMSE*	R^2
4	0.34	-0.90
8	0.34	-0.80
16	0.03	0.88
32	0.04	0.95
64	0.03	0.94
128	0.02	0.95
256	0.04	-0.25
512	0.03	0.83

Table S50. Optimisation of the optimiser in GRU algorithm for MSD data

optimiser	*RMSE*	R^2
Adam	0.02	0.95
AdamW	0.13	-11.73
RMSprop	0.10	-3.95
SGD	0.64	-384.70

Table S51. Optimisation of the data percentage in GRU algorithm for MSD data

data percentage	*RMSE*	R^2
10	0.08	0.87
20	0.10	0.69
30	0.05	0.88
40	0.21	-4.94
50	0.17	-6.16
60	0.04	0.76
70	0.02	0.95
80	0.09	-0.24

Table S52. Optimisation of the number of layers in LSTM algorithm for MSD data

number of layers	*RMSE*	R^2
1	0.02	0.93
2	0.03	0.76
3	0.40	-162.67
4	0.44	-193.15
5	0.66	-408.76
6	0.66	-403.69
7	0.67	-428.54
8	0.67	-422.99
9	0.34	-1.63
10	0.66	-407.50

Table S53. Optimisation of the hidden size in LSTM algorithm for MSD data

hidden size	*RMSE*	R^2
2	0.49	-199.68
4	0.14	-17.62
8	0.07	-3.81
16	0.04	-0.50
32	0.06	-1.55
64	0.02	0.94
128	0.02	0.93
256	0.05	-1.56
512	0.07	-2.13
1024	0.40	-1.61

Table S54. Optimisation of the learning rate in LSTM algorithm for MSD data

learning rate	*RMSE*	R^2
1.00×10^{-06}	0.77	-600.57
1.00×10^{-05}	0.80	-656.58
1.00×10^{-04}	0.13	-7.39
2.00×10^{-04}	0.06	-1.42
3.00×10^{-04}	0.02	0.72
4.00×10^{-04}	0.11	-11.01
5.00×10^{-04}	0.04	0.78
6.00×10^{-04}	0.08	0.86
7.00×10^{-04}	0.06	-1.47
8.00×10^{-04}	0.04	0.53
9.00×10^{-04}	0.03	0.75
1.00×10^{-03}	0.02	0.94
2.00×10^{-03}	0.02	0.80
3.00×10^{-03}	0.02	0.93
4.00×10^{-03}	0.02	0.95
5.00×10^{-03}	0.03	0.39
6.00×10^{-03}	0.03	0.83
7.00×10^{-03}	0.03	0.26
8.00×10^{-03}	0.06	-0.15
9.00×10^{-03}	0.02	0.96
1.00×10^{-02}	0.06	-1.92
1.00×10^{-01}	0.35	-1.09

Table S55. Optimisation of the batch size in LSTM algorithm for MSD data

batch size	*RMSE*	R^2
4	0.08	0.55
8	0.03	0.97
16	0.01	0.97
32	0.02	0.95
64	0.07	-2.63
128	0.11	-12.60
256	0.04	0.43
512	0.05	0.91

Table S56. Optimisation of the optimiser in LSTM algorithm for MSD data

optimiser	*RMSE*	R^2
Adam	0.01	0.99
AdamW	0.02	0.99
RMSprop	0.04	0.83
SGD	0.25	-47.01

Table S57. Optimisation of the data percentage in LSTM algorithm for MSD data

data percentage	*RMSE*	R^2
10	0.11	0.75
20	0.05	0.89
30	0.02	0.95
40	0.01	0.99
50	0.01	0.99
60	0.03	0.99
70	0.01	0.99
80	0.02	0.99

Table S58. Optimisation of the number of layers in LAG-LLAMA algorithm for MSD data

number of layers	*RMSE*	R^2
1	0.04	0.58
2	0.14	-2.32
3	0.59	-292.20
4	0.18	-0.40
5	0.20	-0.67
6	0.57	-255.77
7	0.56	-240.03
8	0.59	-297.27
9	0.28	-33.48
10	0.55	-233.14
11	0.50	-157.26
12	0.55	-236.37
13	0.40	-28.32
14	0.57	-265.50
15	0.55	-228.06
16	0.57	-255.06

Table S59. Optimisation of the d_model in LAG-LLAMA algorithm for MSD data

d_model	*RMSE*	R^2
4	0.19	-41.54
8	0.05	0.19
16	0.04	0.60
32	0.03	0.68
64	0.04	0.58
128	0.04	0.55
256	0.11	-1.72
512	0.22	0.05
1024	0.61	-69.10

Table S60. Optimisation of the n_heads in LAG-LLAMA algorithm for MSD data

n_heads	*RMSE*	R^2
1	0.03	0.76
2	0.06	-0.01
4	0.06	0.44
8	0.04	0.61
16	0.03	0.64

Table S61. Optimisation of the dropout in LAG-LLAMA algorithm for MSD data

dropout	*RMSE*	R^2
0.01	0.05	0.65
0.02	0.08	-1.59
0.03	0.04	0.70
0.04	0.06	0.07
0.05	0.02	0.88
0.06	0.03	0.80
0.07	0.08	-0.65
0.08	0.04	0.69
0.09	0.07	-0.84
0.1	0.03	0.76
0.11	0.07	-0.71
0.12	0.06	0.29
0.13	0.07	-0.55
0.14	0.07	-1.00
0.15	0.05	0.28
0.16	0.05	0.18
0.17	0.05	0.14
0.18	0.05	0.36
0.19	0.05	0.67
0.2	0.04	0.85

Table S62. Optimisation of the learning rate in LAG-LLAMA algorithm for MSD data

learning rate	*RMSE*	R^2
1.00×10^{-06}	0.45	-138.43
1.00×10^{-05}	0.34	-124.79
1.00×10^{-04}	0.05	-1.38
2.00×10^{-04}	0.03	0.65
3.00×10^{-04}	0.02	0.73
4.00×10^{-04}	0.03	0.51
5.00×10^{-04}	0.03	0.89
6.00×10^{-04}	0.03	0.79
7.00×10^{-04}	0.02	0.90
8.00×10^{-04}	0.04	0.63
9.00×10^{-04}	0.05	-0.06
1.00×10^{-03}	0.02	0.88
2.00×10^{-03}	0.06	0.32
3.00×10^{-03}	0.17	-4.74
4.00×10^{-03}	0.09	0.23
5.00×10^{-03}	0.06	0.28
6.00×10^{-03}	0.66	-407.07
7.00×10^{-03}	0.61	-324.27
8.00×10^{-03}	0.31	-3.08
9.00×10^{-03}	0.62	-336.23
1.00×10^{-02}	0.66	-411.29
1.00×10^{-01}	0.35	-5.91

Table S63. Optimisation of the batch size in LAG-LLAMA algorithm for MSD data

batch size	*RMSE*	R^2
4	0.12	0.13
8	0.05	0.44
16	0.05	0.51
32	0.02	0.90
64	0.04	0.72
128	0.04	0.14
256	0.05	-1.47
512	0.06	-0.73

Table S64. Optimisation of the optimiser in LAG-LLAMA algorithm for MSD data

optimiser	*RMSE*	R^2
Adam	0.02	0.90
AdamW	0.01	0.93
RMSprop	0.05	0.52
SGD	0.05	0.48

Table S65. Optimisation of the data percentage in LAG-LLAMA algorithm for MSD data

data percentage	*RMSE*	R^2
10	0.02	0.96
20	0.10	0.64
30	0.11	0.29
40	0.10	-0.03
50	0.07	-0.79
60	0.03	0.76
70	0.01	0.98
80	0.03	0.19

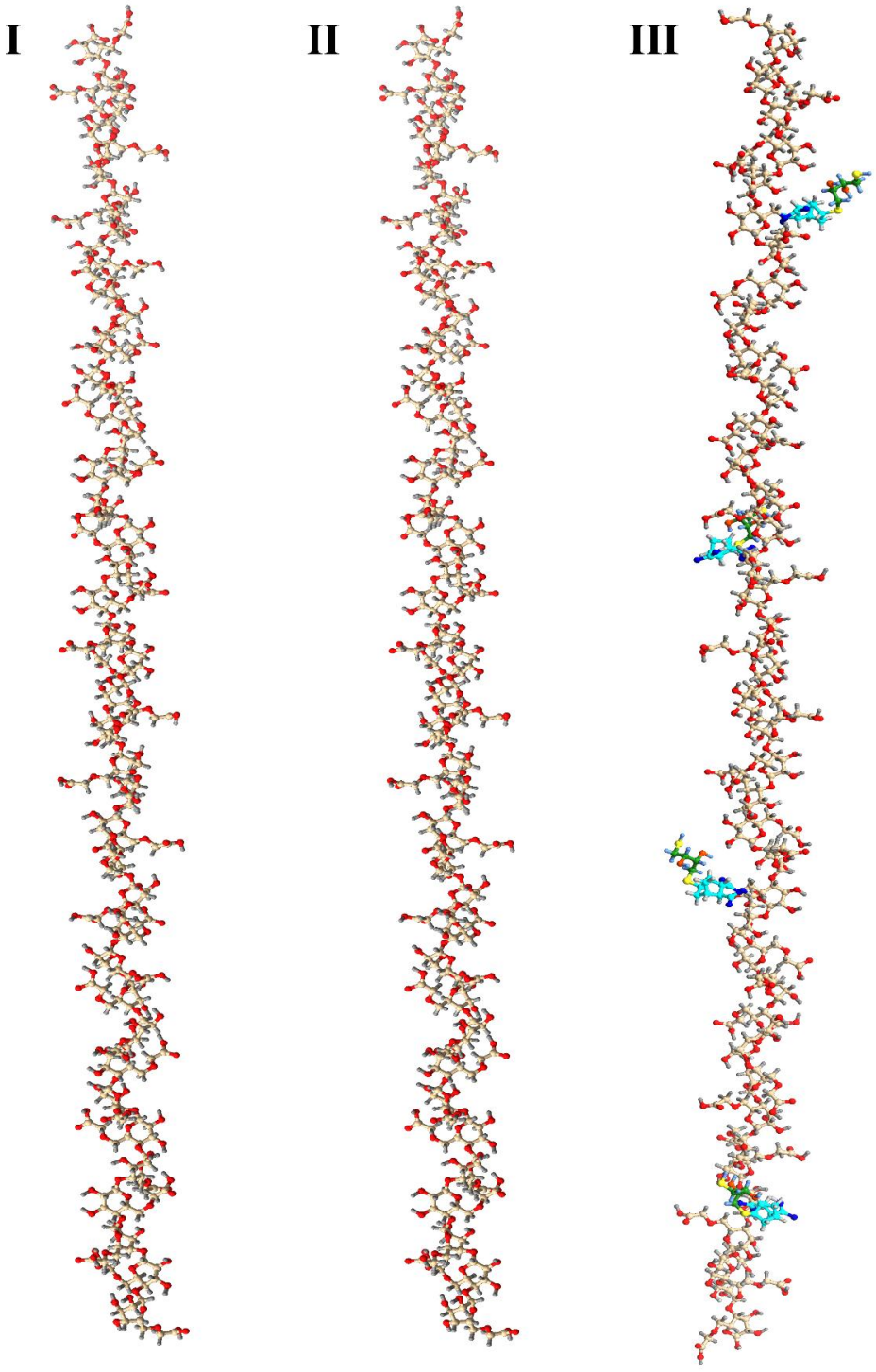


Figure S1. The initial configuration of the polymer chains I. CMC, II. CMC-N, and III. CMC-NT

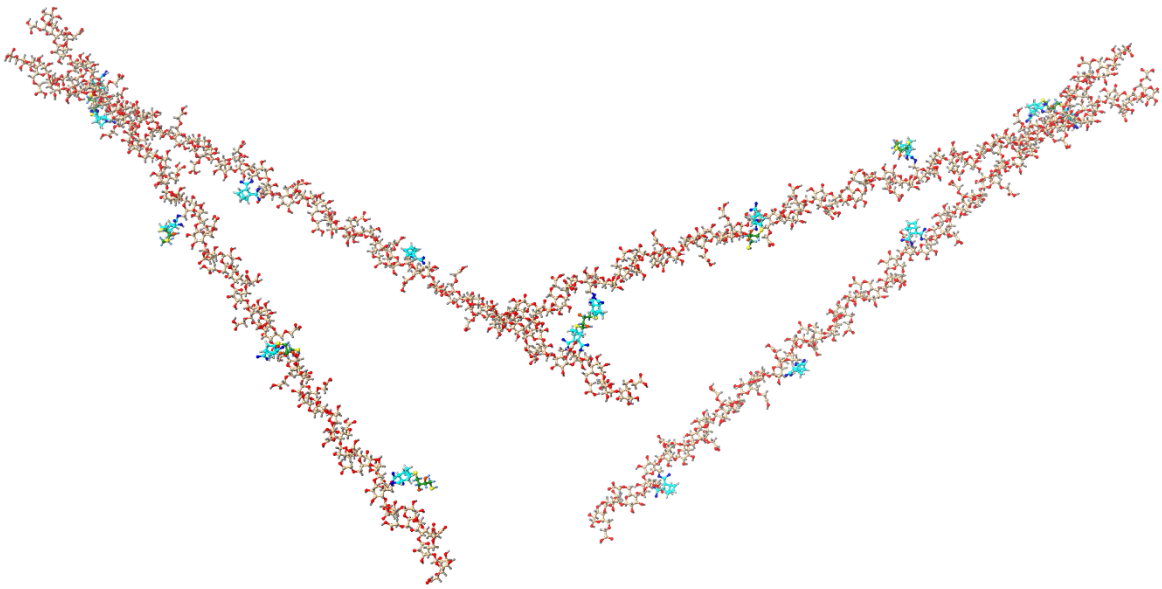


Figure S2. The initial configuration of the CMC-NTC polymer chains

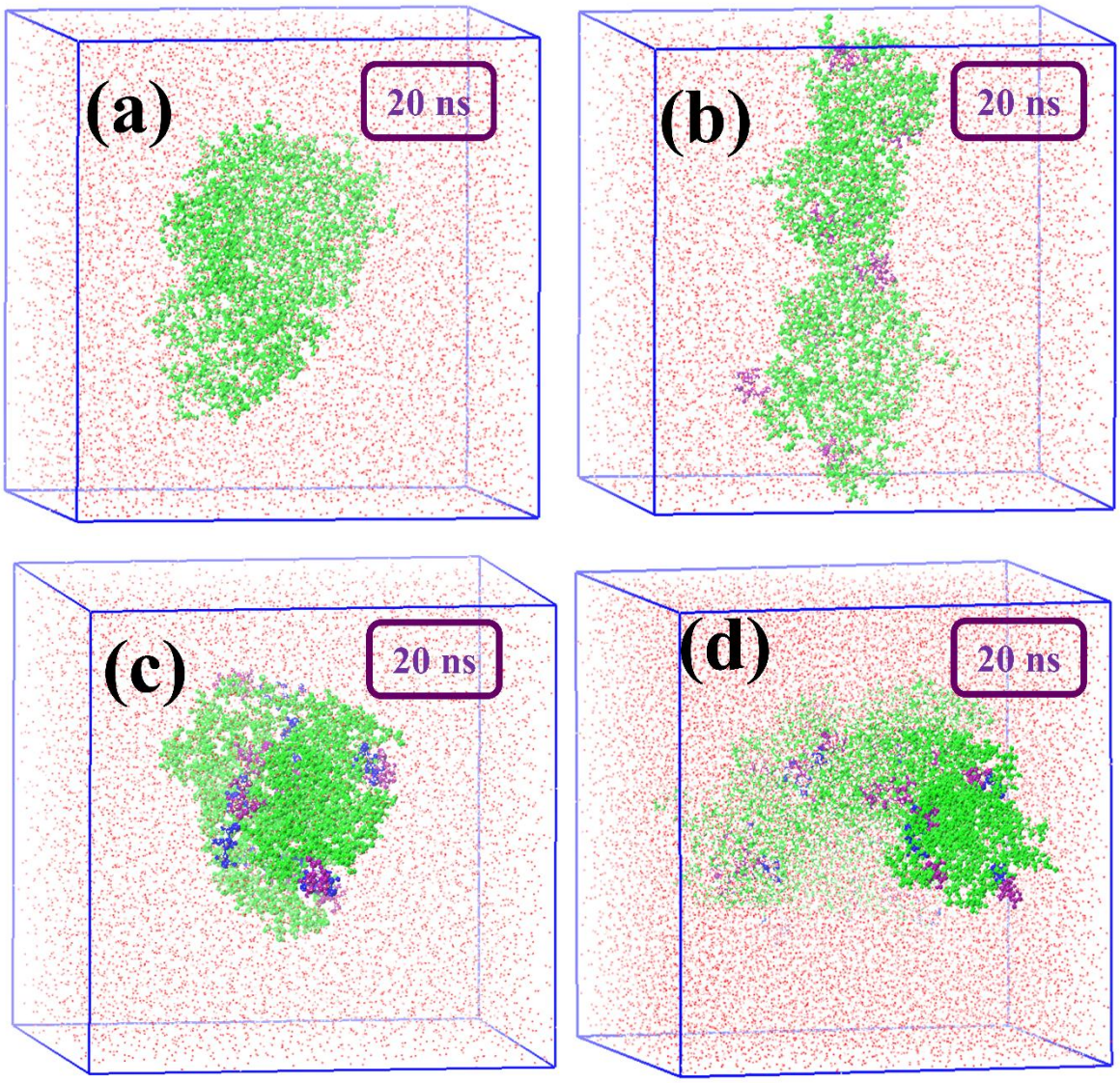


Figure S3**.** The final configuration of gel (a) CMC, (b) CMC–N, (c) CMC–NT, and (d) CMC–NTC

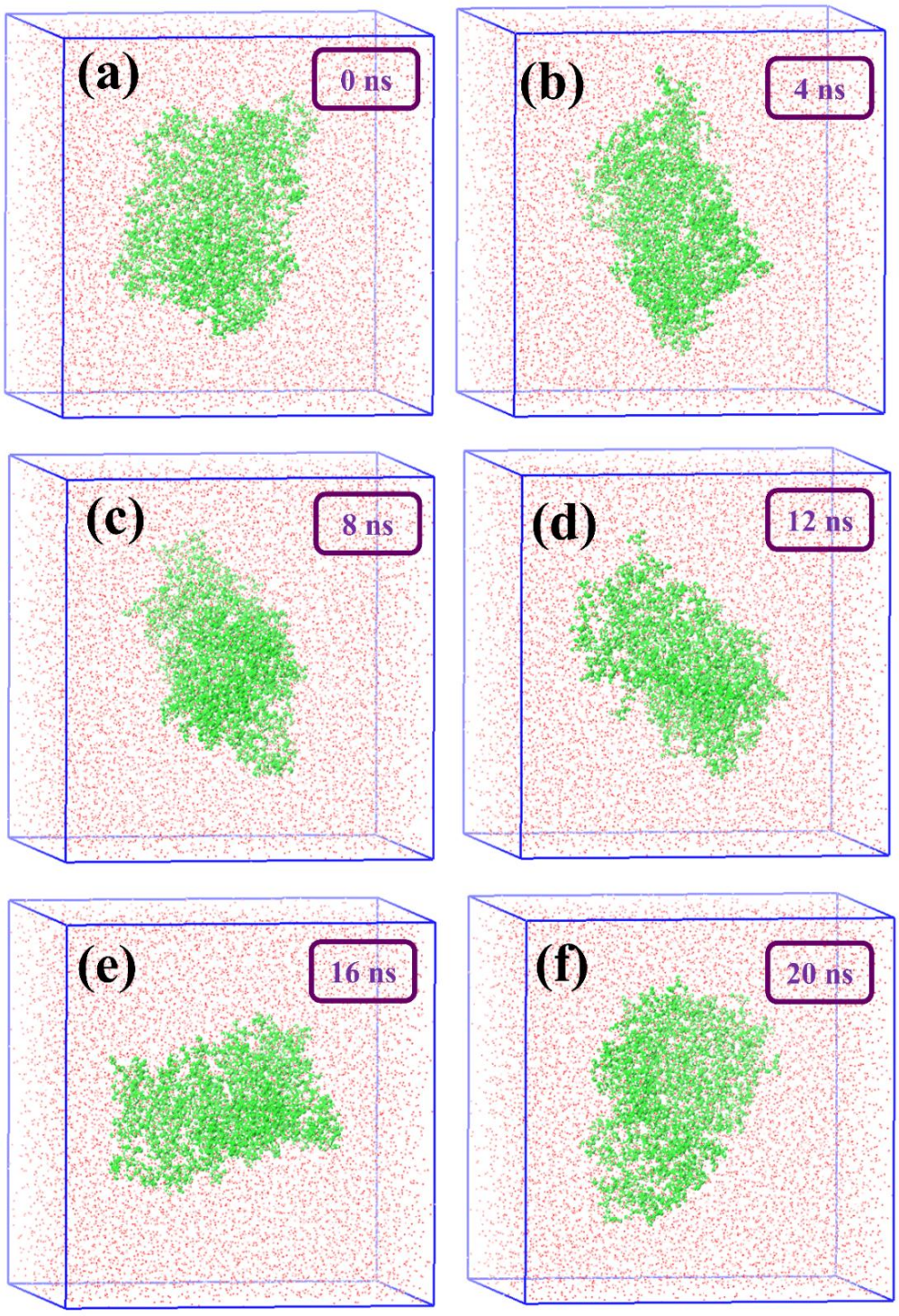


Figure S4. The chronological configuration of the CMC gel (a) 0 ns, (b) 4 ns, (c) 8 ns, (d) 12 ns, (e) 16 ns, and (f) 20 ns

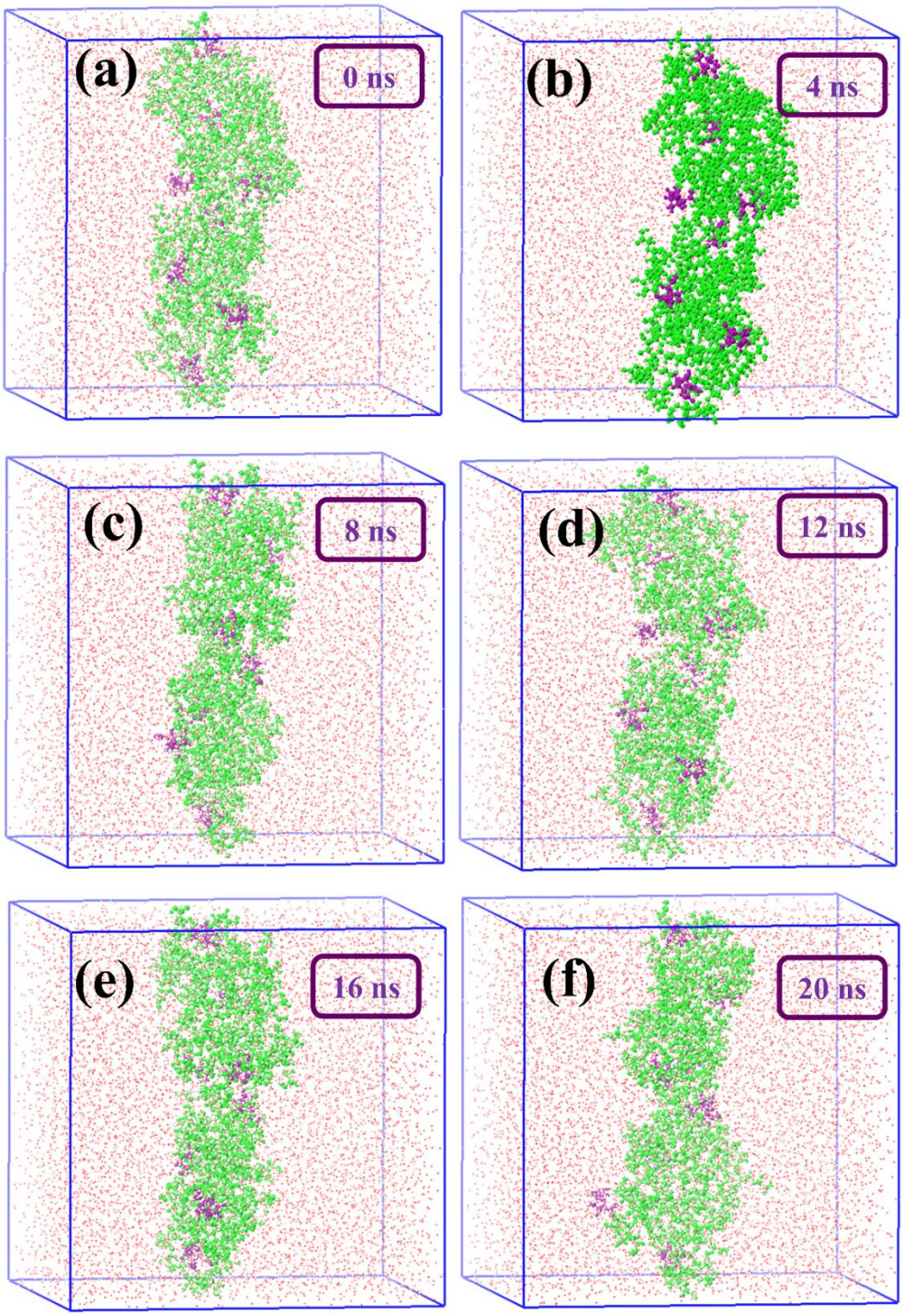


Figure S5. The chronological configuration of the CMC-N gel (a) 0 ns, (b) 4 ns, (c) 8 ns, (d) 12 ns, (e) 16 ns, and (f) 20 ns

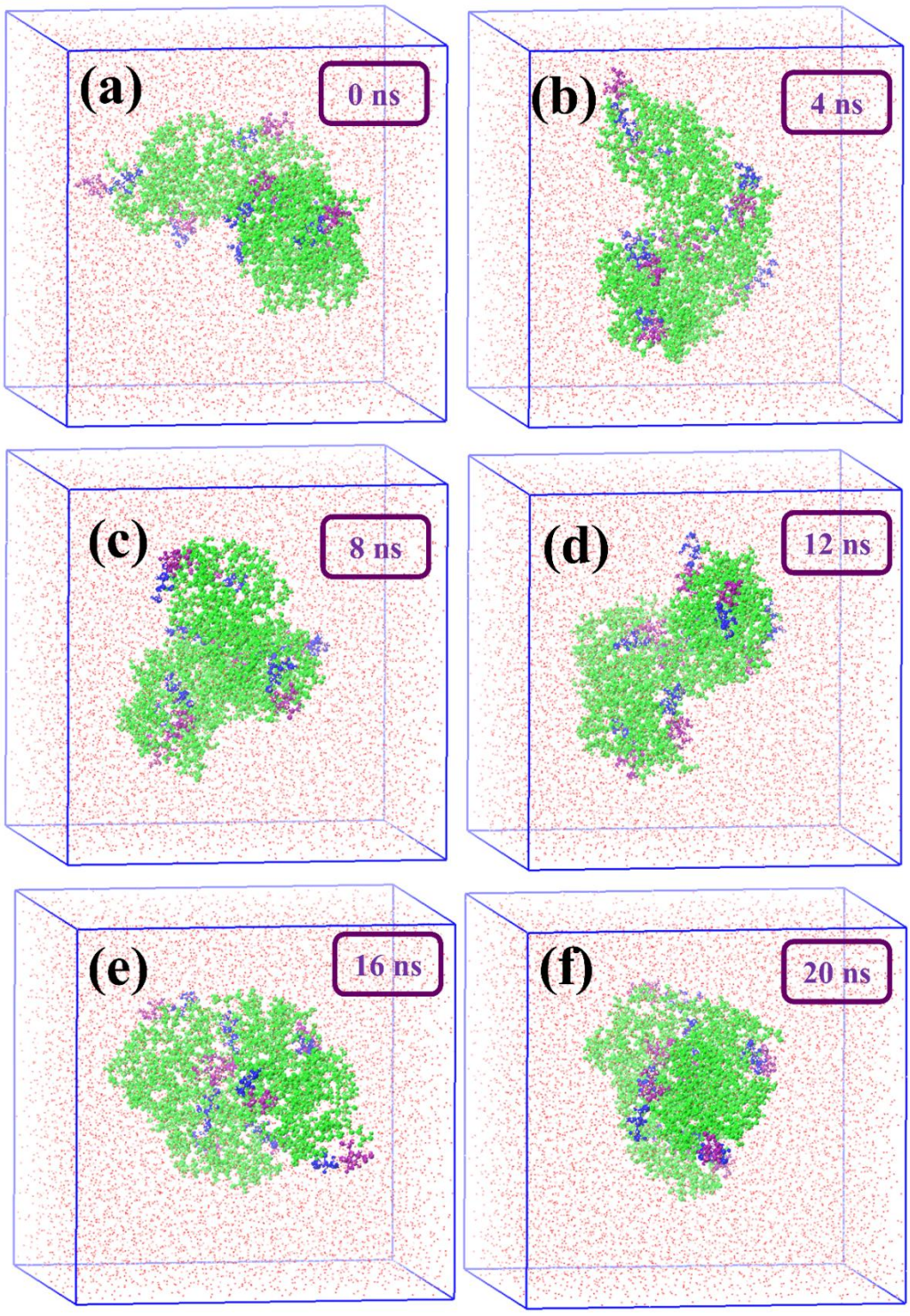


Figure S6. The chronological configuration of the CMC-NT gel (a) 0 ns, (b) 4 ns, (c) 8 ns, (d) 12 ns, (e) 16 ns, and (f) 20 ns

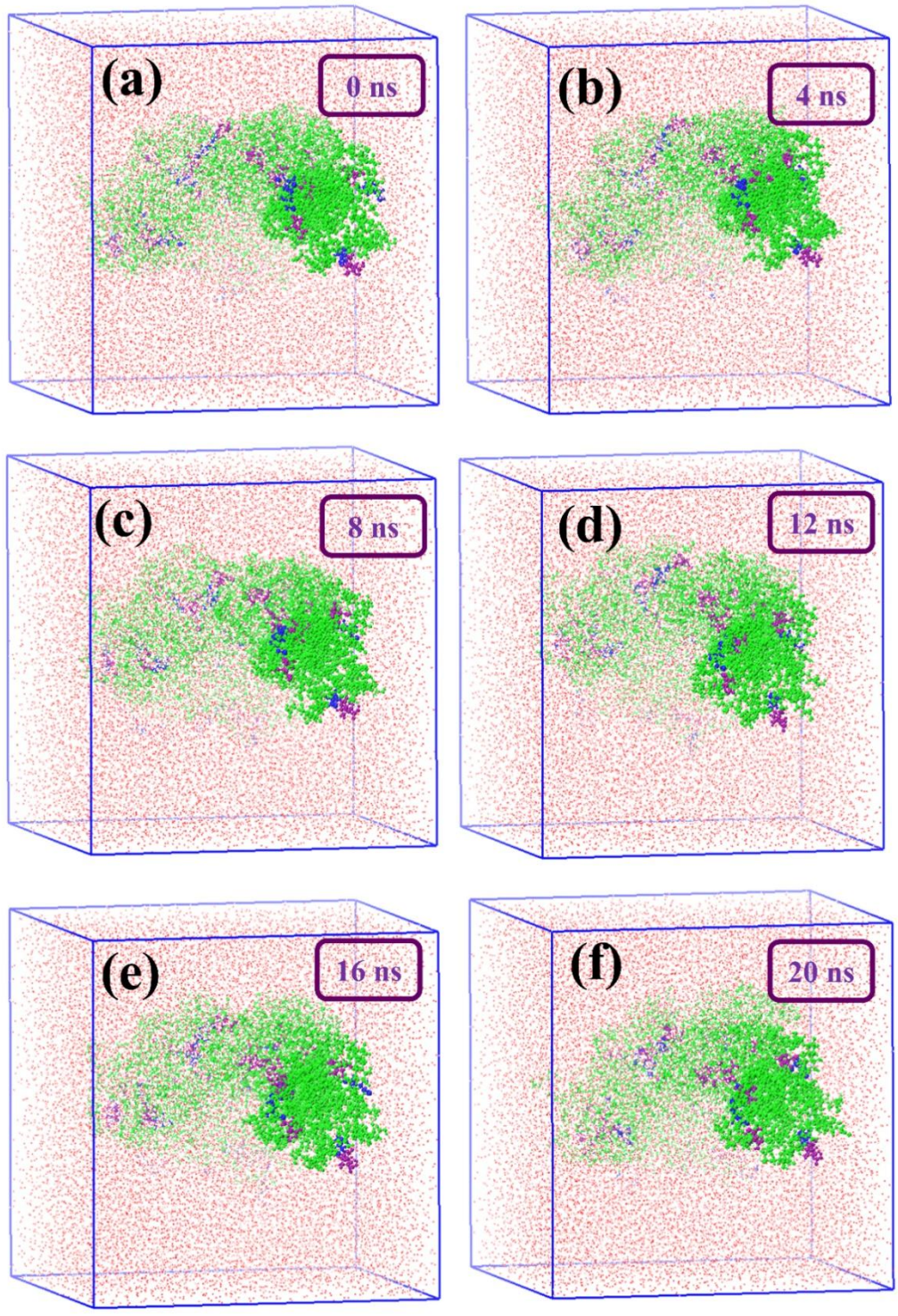


Figure S7. The chronological configuration of the CMC-NTC gel (a) 0 ns, (b) 4 ns, (c) 8 ns, (d) 12 ns, (e) 16 ns, and (f) 20 ns

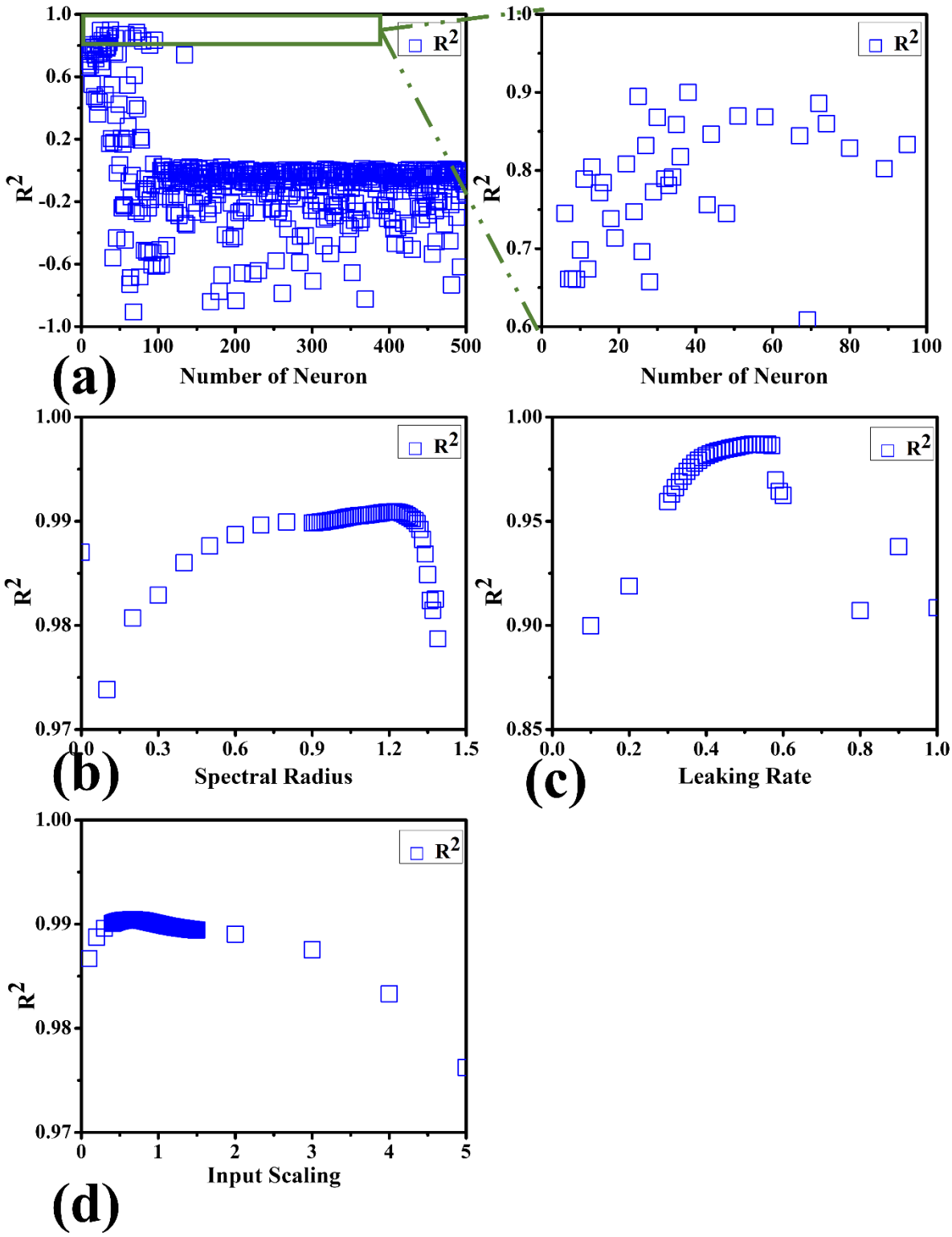


Figure S8. Optimisation ESN parameters for Hbond data (a) Number of neuron (b) spectral radius (c) leaking rate (d) input scaling

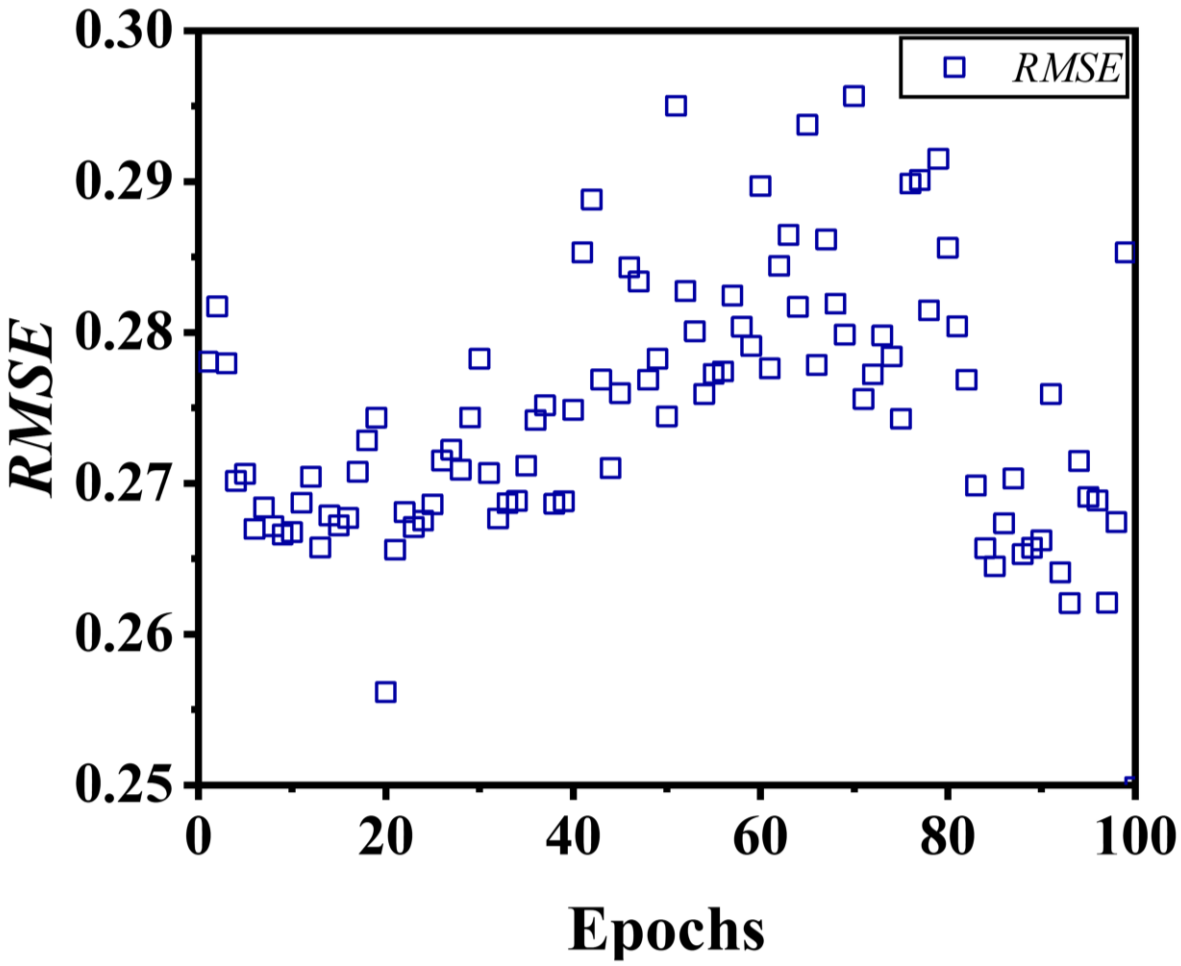


Figure S9. Optimising the Epochs in the GRU algorithm for the Hbonds data

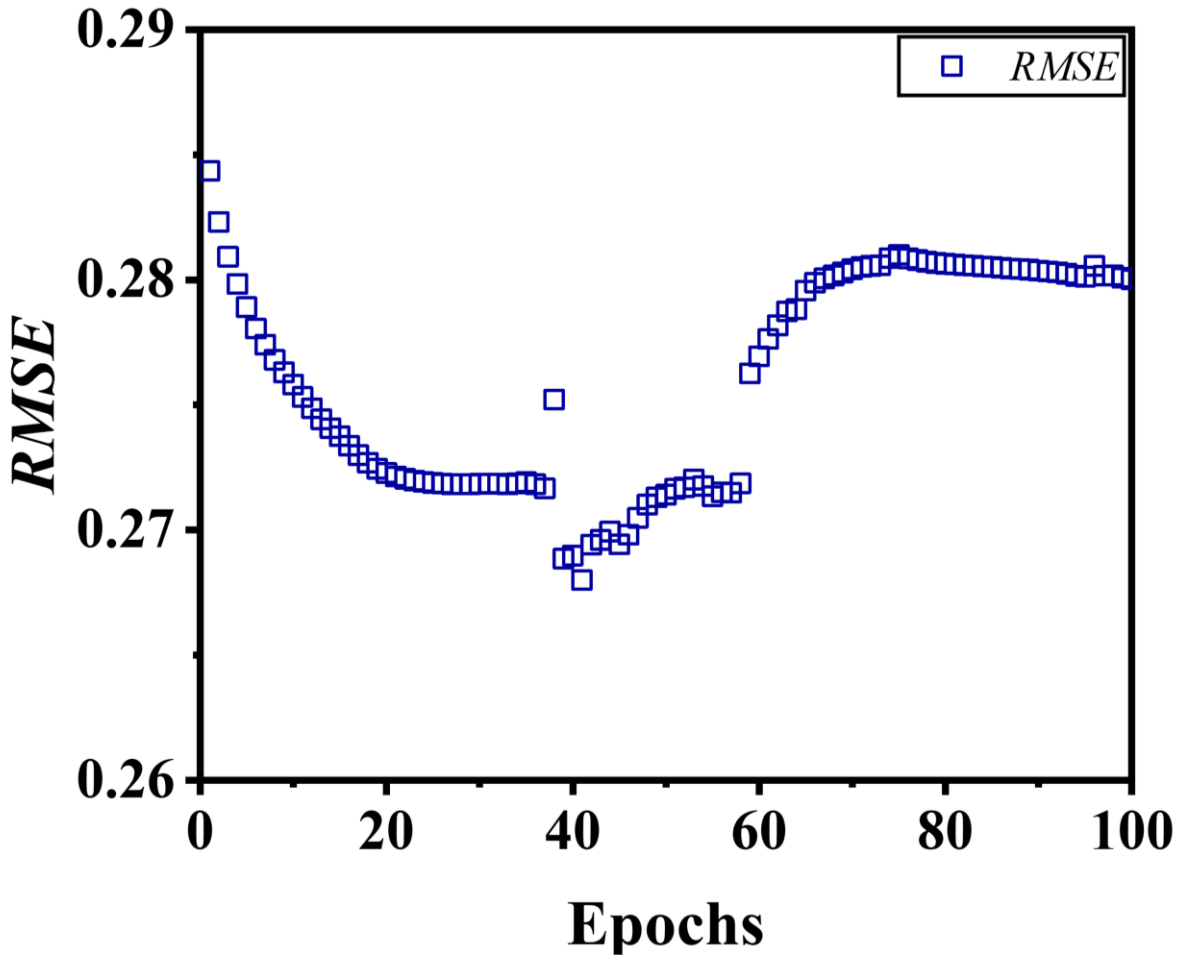


Figure S10. Optimising the Epochs in the LSTM algorithm for the Hbonds data

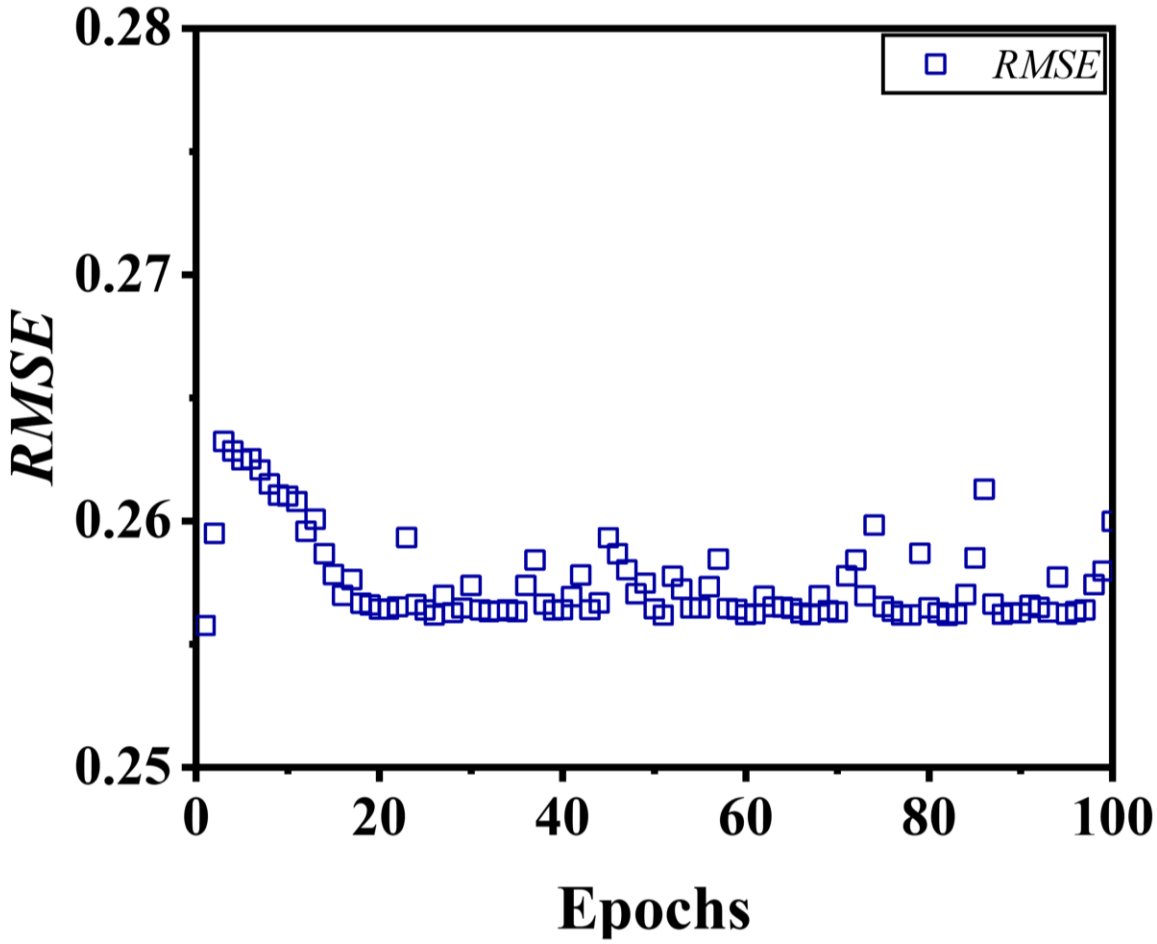


Figure S11. Optimising the Epochs in the LAG-LLAMA algorithm for the Hbonds data

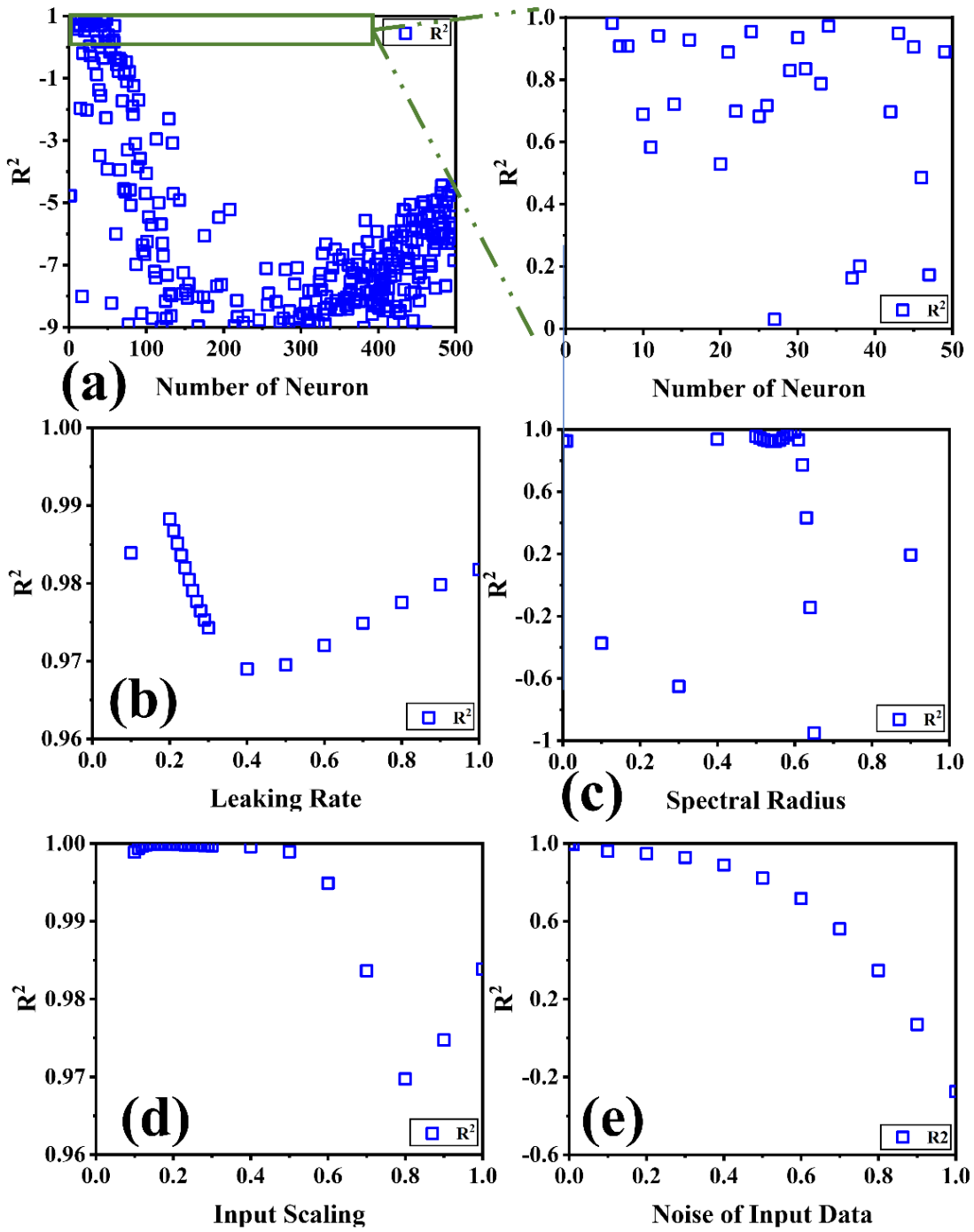


Figure S12. Optimisation ESN parameters for MSD data (a) Number of neuron (b) leaking rate (c) spectral radius (d) input scaling (e) noise of input data

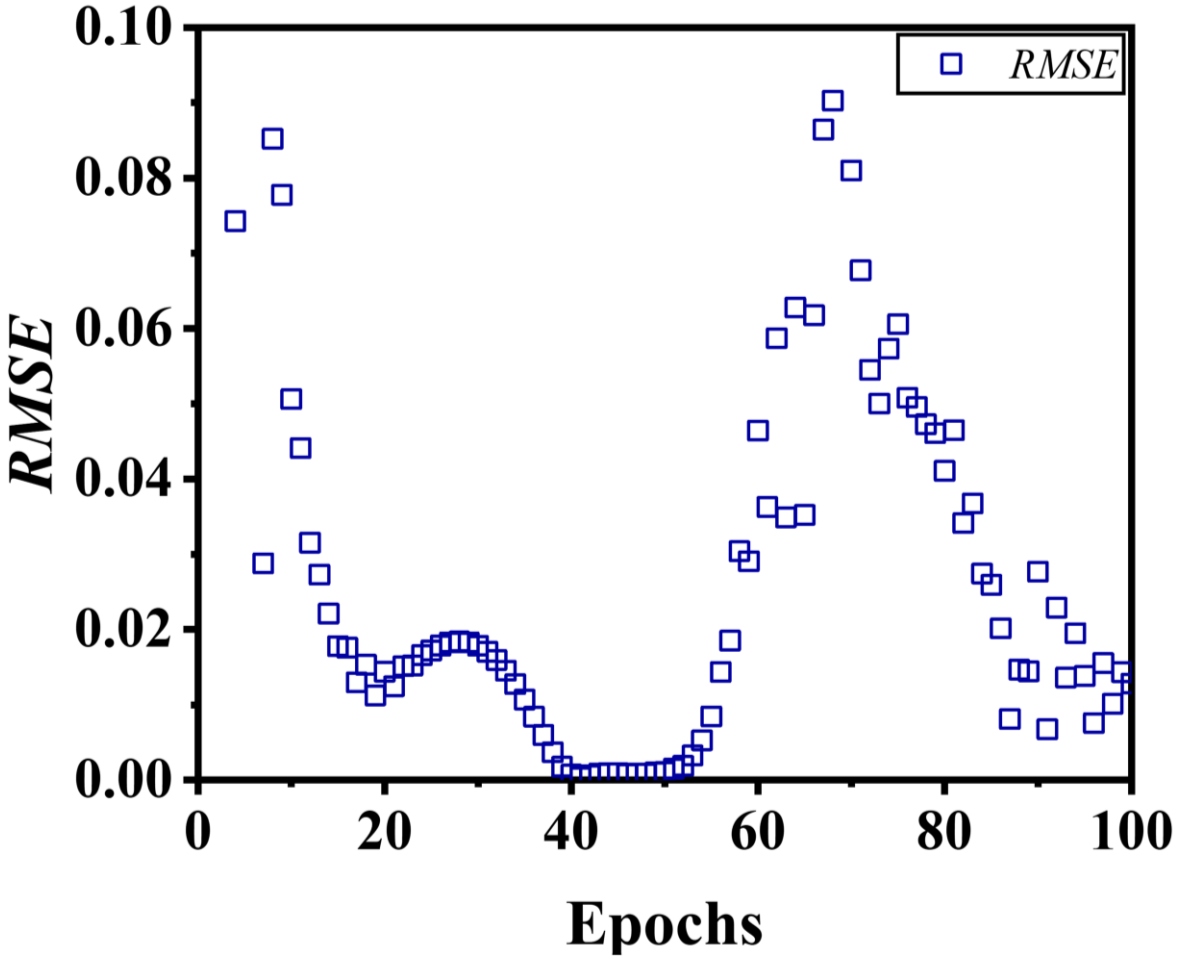


Figure S13. Optimising the Epochs in the GRU algorithm for the MSD data

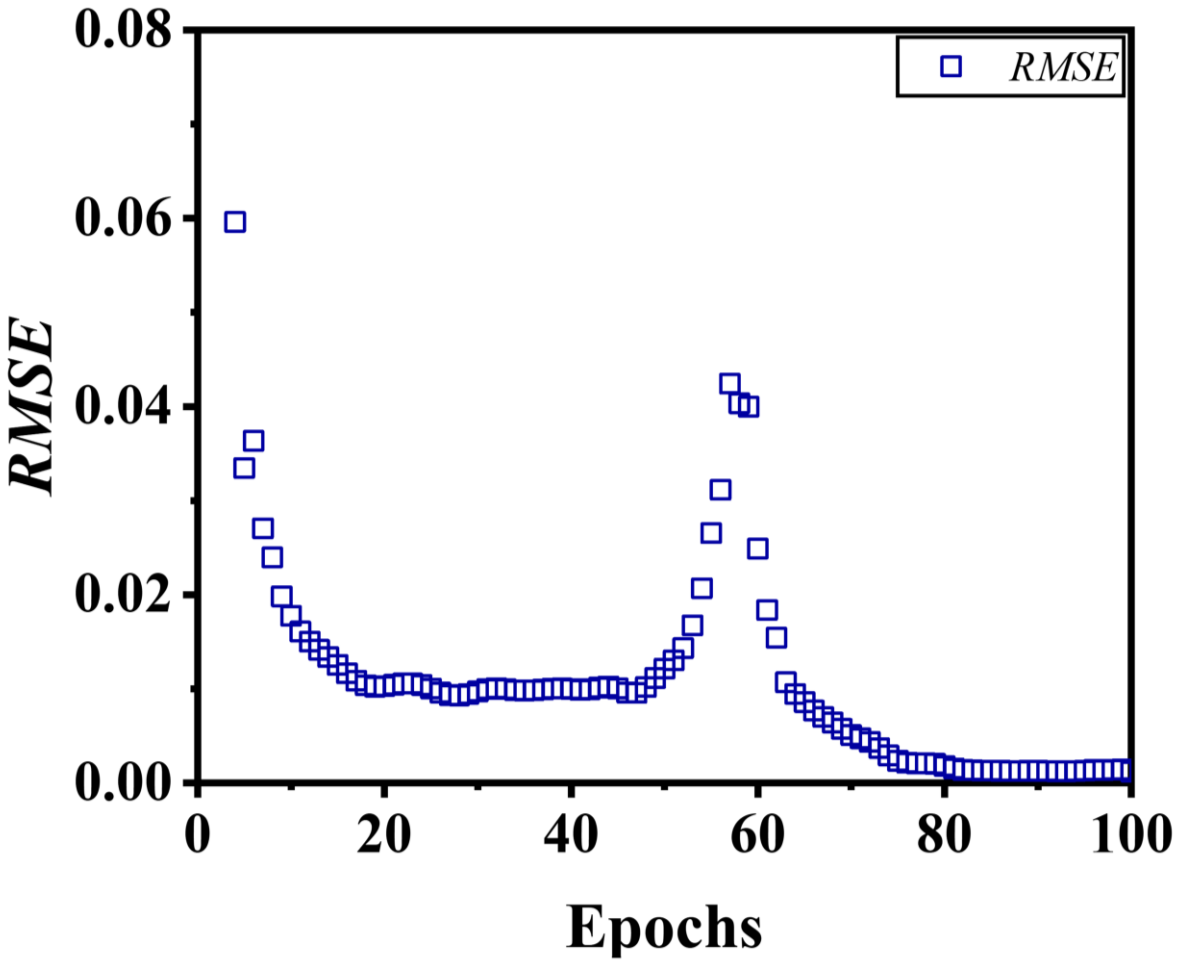


Figure S14. Optimising the Epochs in the LSTM algorithm for the MSD data

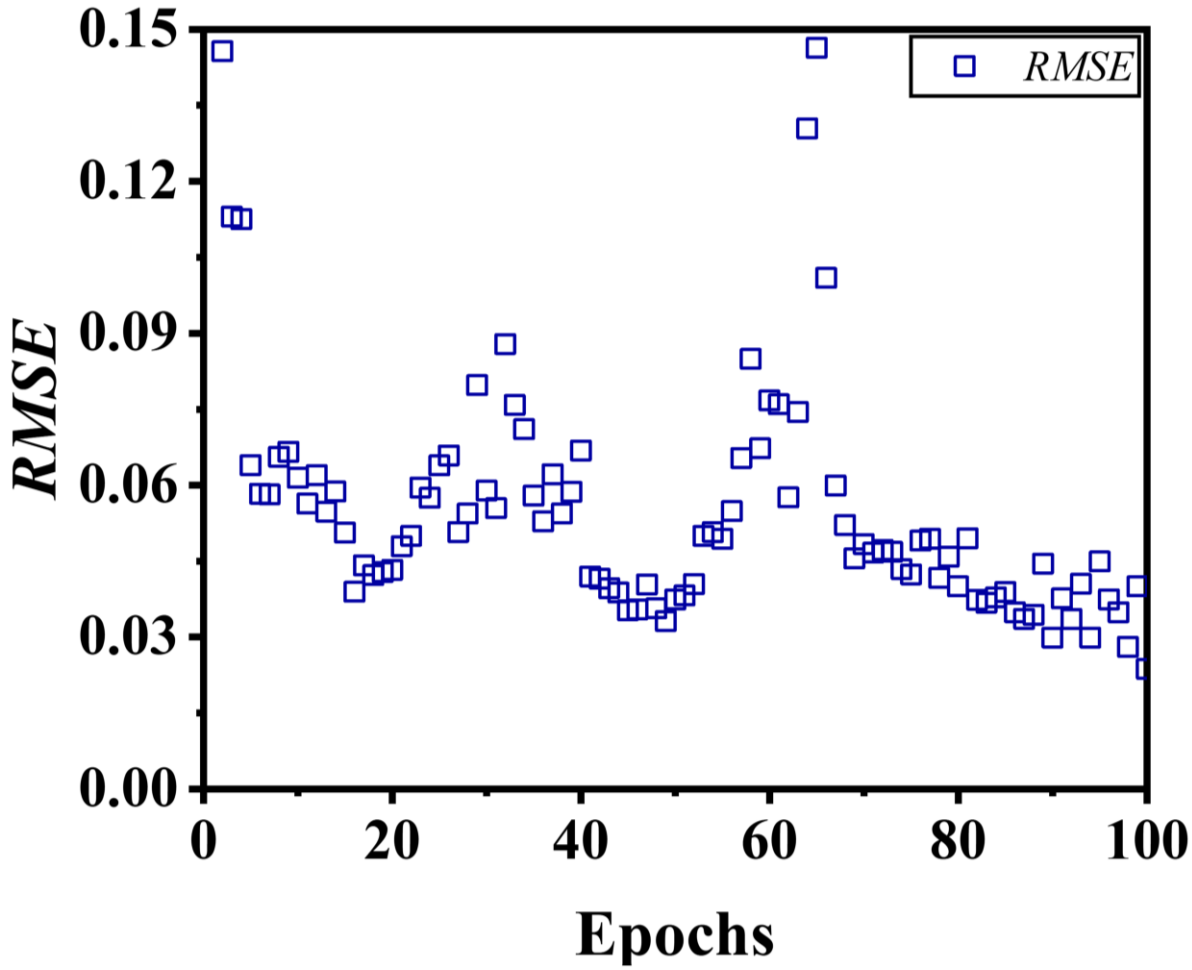
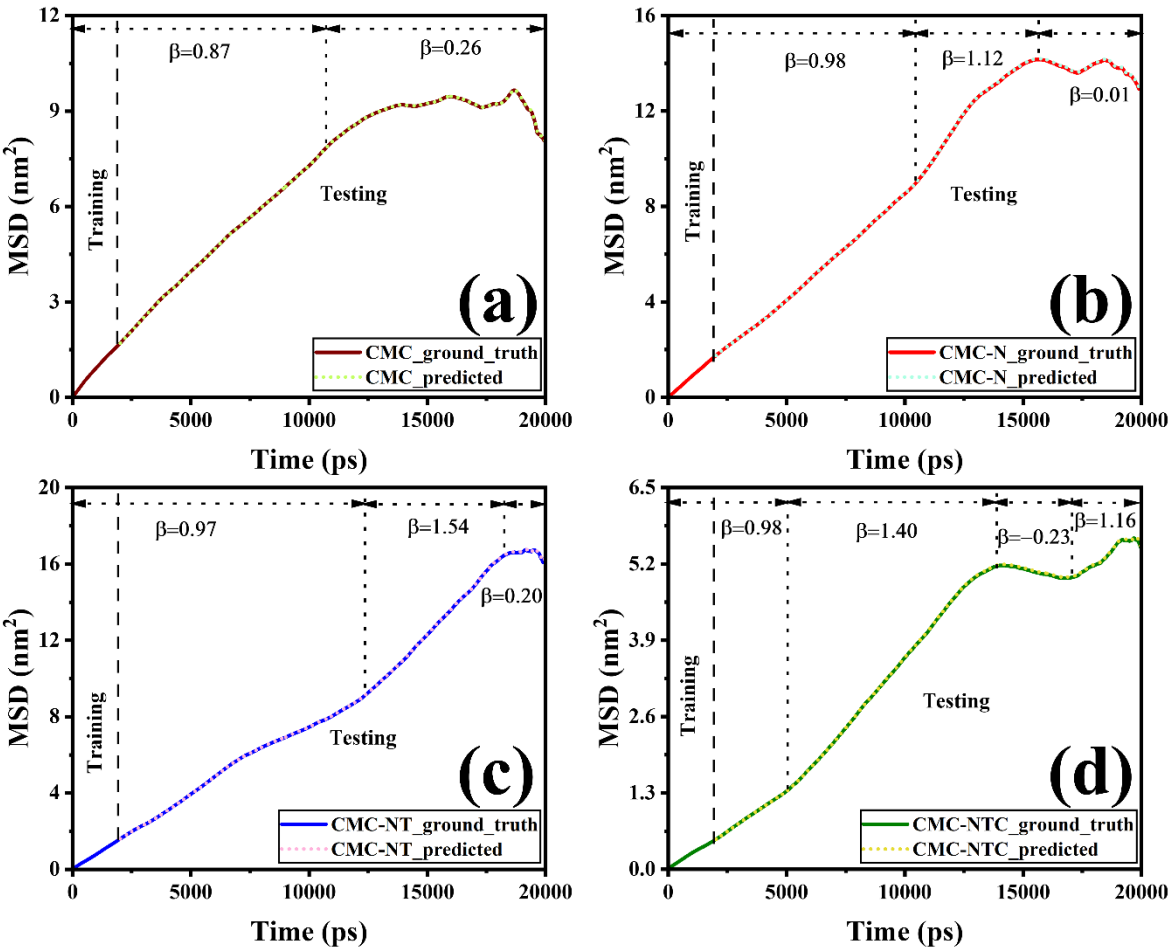


Figure S15. Optimising the Epochs in the LAG-LLAMA algorithm for the MSD data


(a)
β=0.87
β=0.26
Training
Testing
MSD (nm²)
Time (ps)
CMC_ground_truth
CMC_predicted
(b)
β=0.98
β=1.12
β=0.01
Training
Testing
MSD (nm²)
Time (ps)
CMC-N_ground_truth
CMC-N_predicted
(c)
β=0.97
β=1.54
β=0.20
Training
Testing
MSD (nm²)
Time (ps)
CMC-NT_ground_truth
CMC-NT_predicted
(d)
β=0.98
β=1.40
β=−0.23
β=1.16
Training
Testing
MSD (nm²)
Time (ps)
CMC-NTC_ground_truth
CMC-NTC_predicted

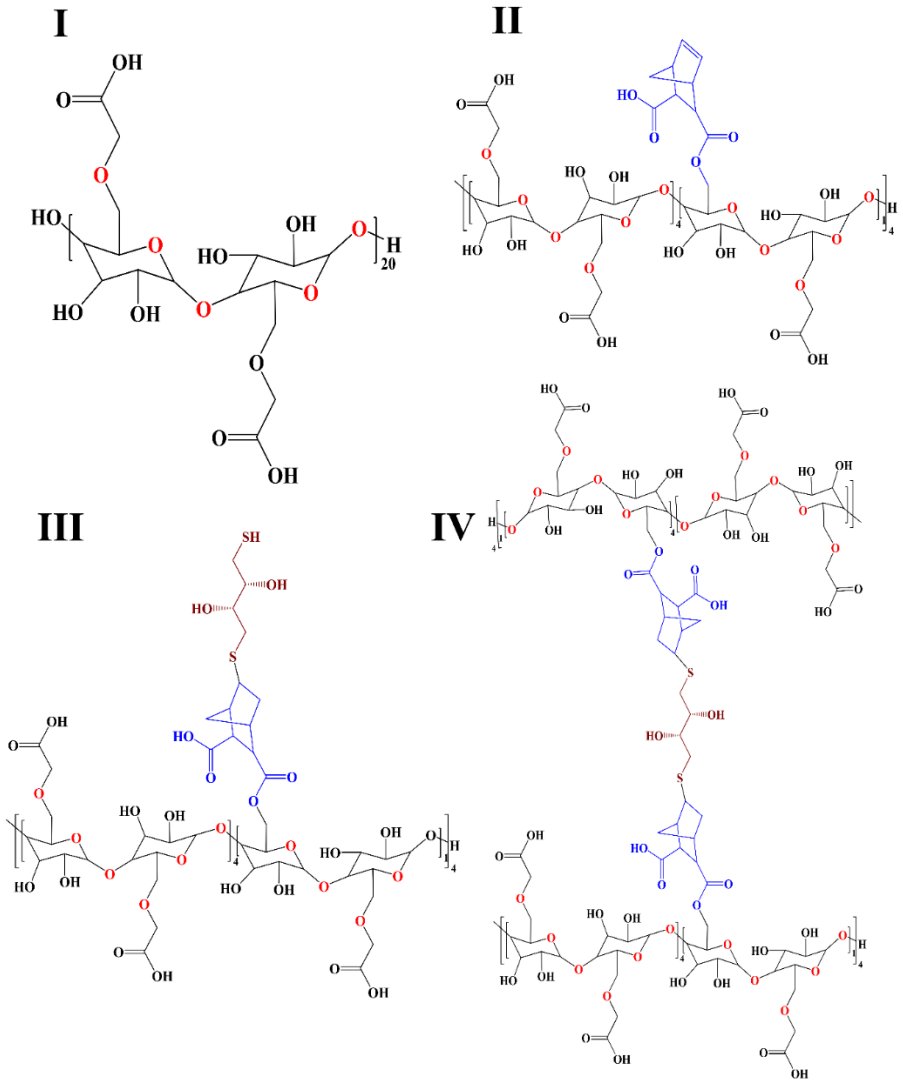
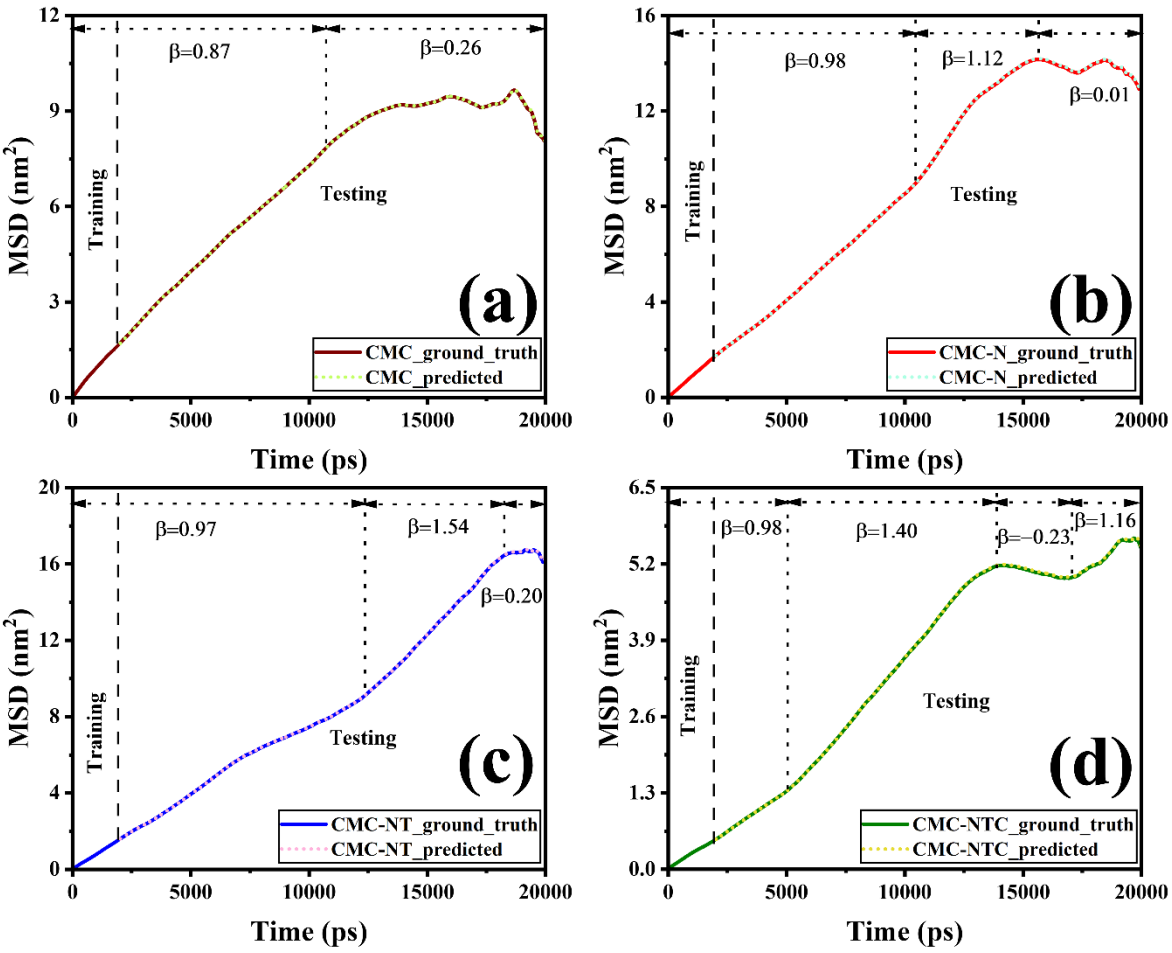


Figure S16. The MSD curve of ether moieties in the polymer backbone. (a) illustrate the MSD curve of CMC, (b) illustrate the MSD curve of CMC–N, (c) illustrate the MSD curve of CMC–NT polymer chain, and (d) represents the MSD curve of CMC–NTC polymer chain, I. schematic diagram of CMC, II. schematic diagram of CMC-N, III. schematic diagram of CMC–NT, and IV. schematic diagram of CMC–NTC


(a)
β=0.87
β=0.26
Training
Testing
MSD (nm²)
Time (ps)
CMC_ground_truth
CMC_predicted
(b)
β=0.98
β=1.12
β=0.01
Training
Testing
MSD (nm²)
Time (ps)
CMC-N_ground_truth
CMC-N_predicted
(c)
β=0.97
β=1.54
β=0.20
Training
Testing
MSD (nm²)
Time (ps)
CMC-NT_ground_truth
CMC-NT_predicted
(d)
β=0.98
β=1.40
β=−0.23
β=1.16
Training
Testing
MSD (nm²)
Time (ps)
CMC-NTC_ground_truth
CMC-NTC_predicted

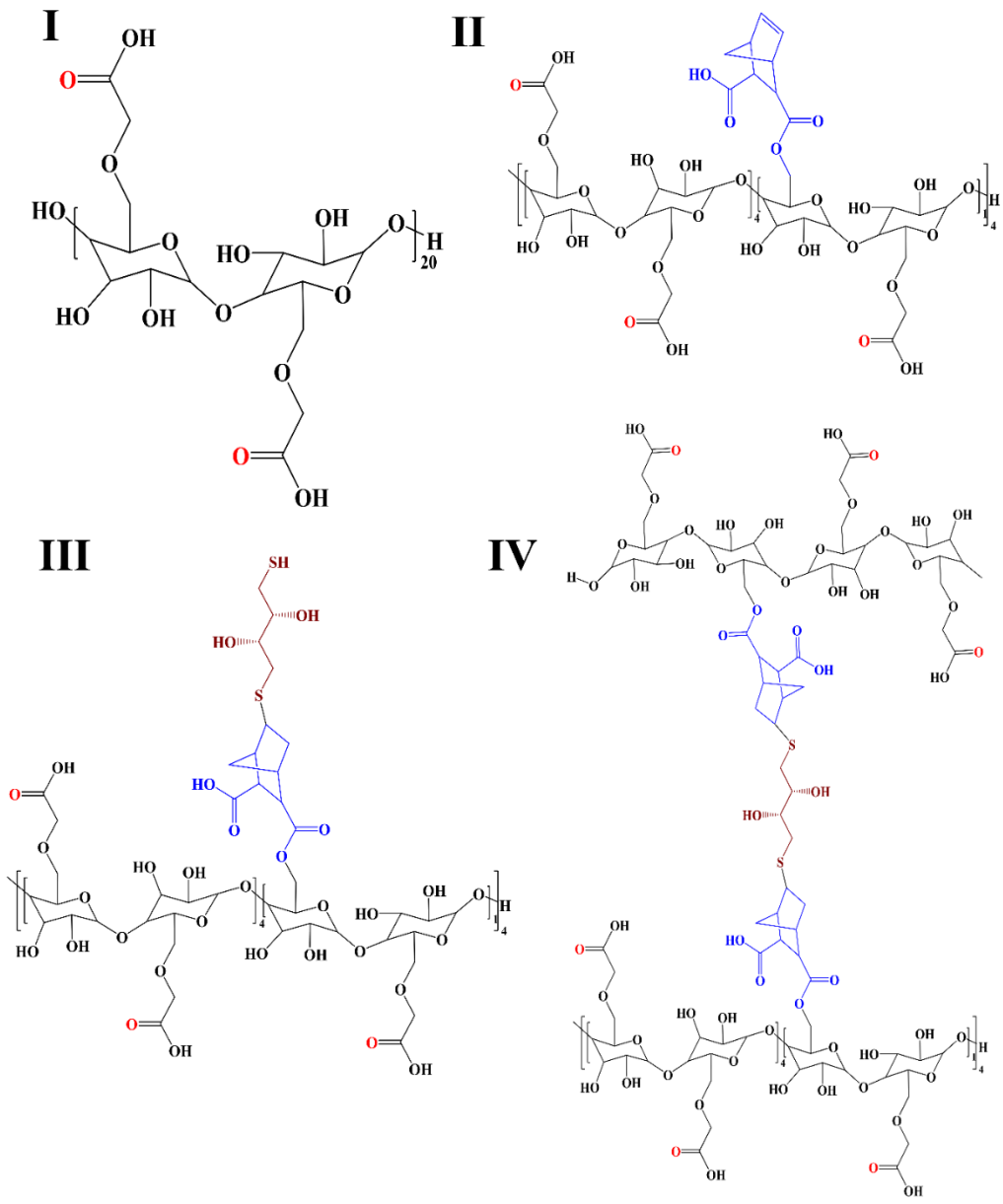
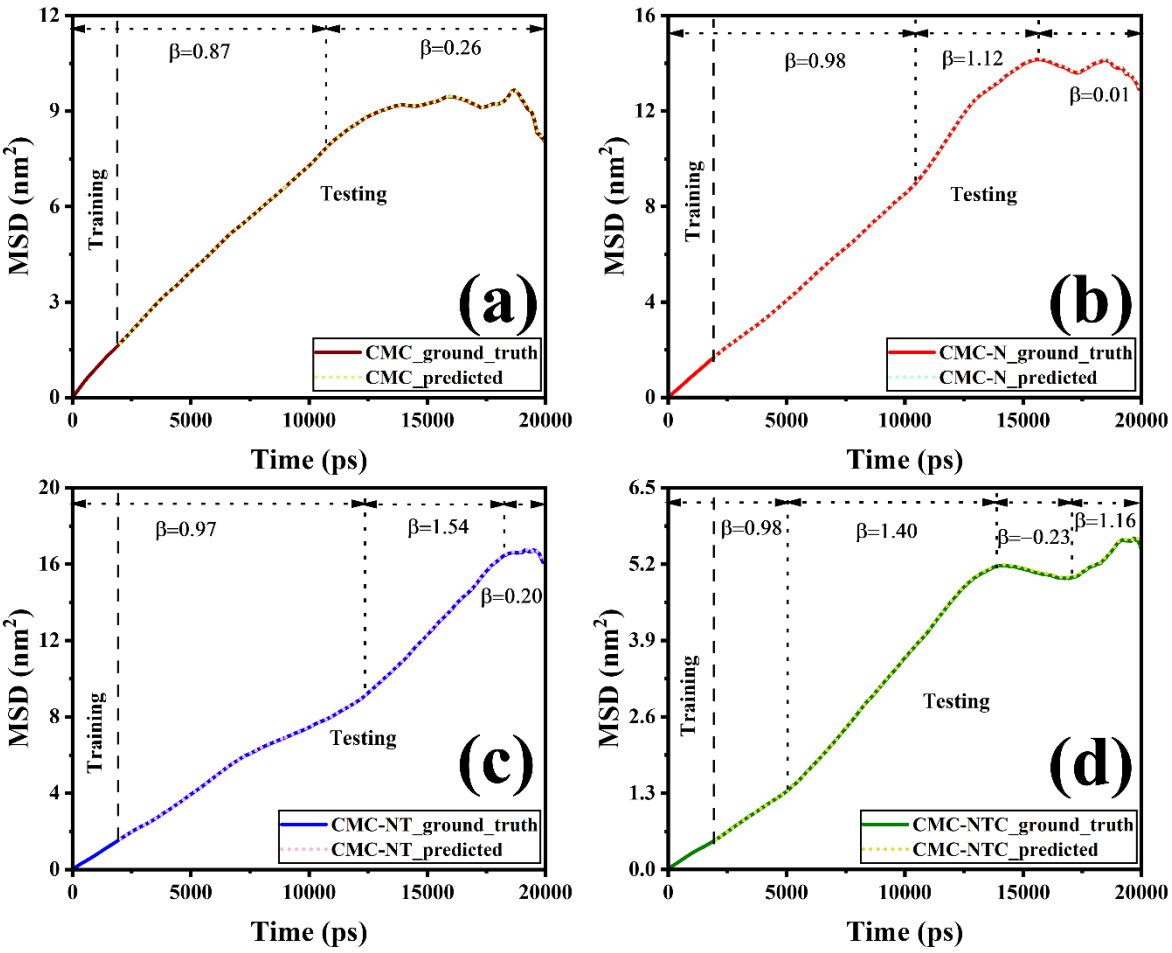


Figure S17. The MSD curve of oxygen atoms of carbonyl moieties in the polymer backbone. (a) illustrate the MSD curve of CMC, (b) illustrate the MSD curve of CMC–N, (c) illustrate the MSD curve of CMC–NT polymer chain, and (d) represents the MSD curve of CMC–NTC polymer chain, I. schematic diagram of CMC, II. schematic diagram of CMC-N, III. schematic diagram of CMC–NT, and IV. schematic diagram of CMC–NTC


(a)
β=0.87
β=0.26
Training
Testing
MSD (nm²)
Time (ps)
CMC_ground_truth
CMC_predicted
(b)
β=0.98
β=1.12
β=0.01
Training
Testing
MSD (nm²)
Time (ps)
CMC-N_ground_truth
CMC-N_predicted
(c)
β=0.97
β=1.54
β=0.20
Training
Testing
MSD (nm²)
Time (ps)
CMC-NT_ground_truth
CMC-NT_predicted
(d)
β=0.98
β=1.40
β=−0.23
β=1.16
Training
Testing
MSD (nm²)
Time (ps)
CMC-NTC_ground_truth
CMC-NTC_predicted

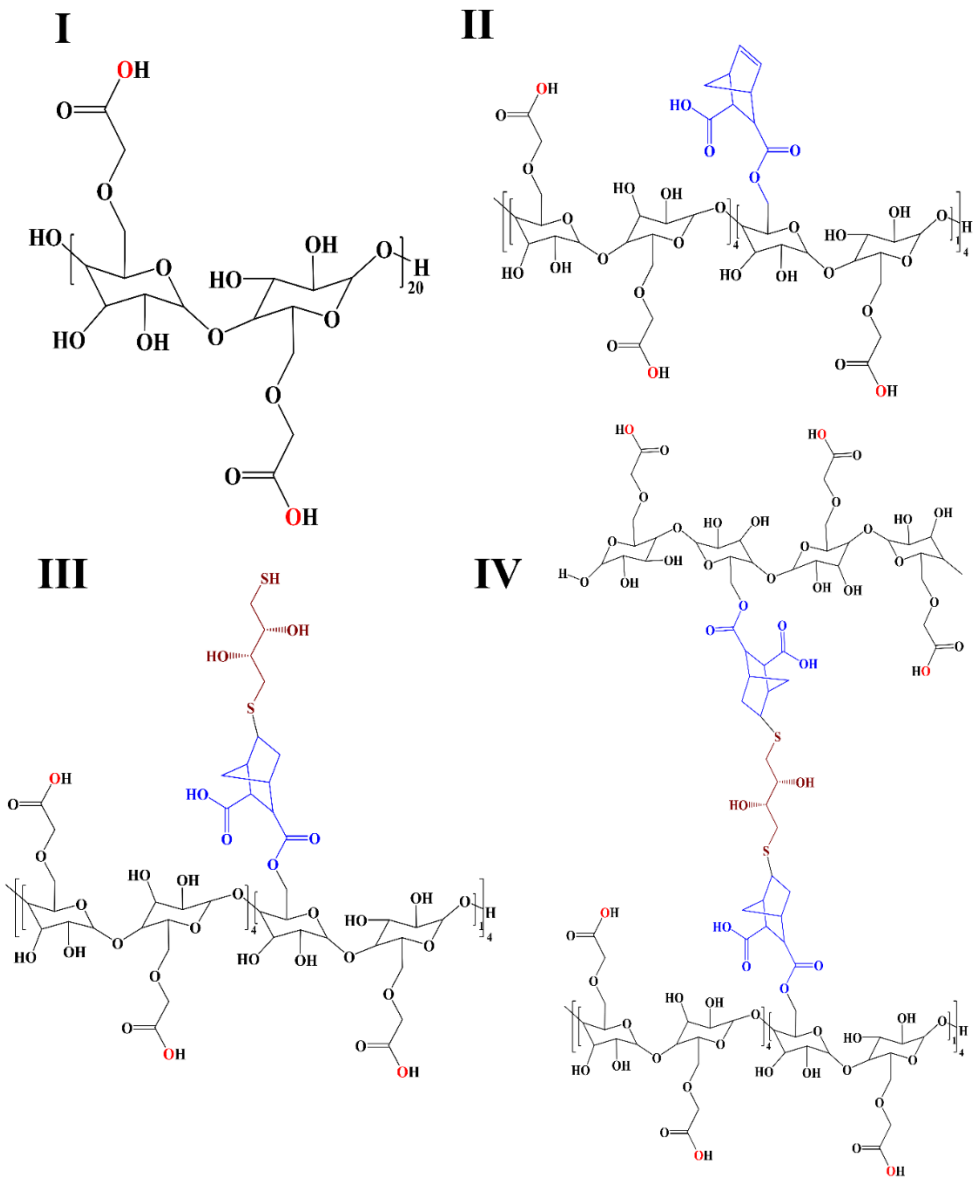
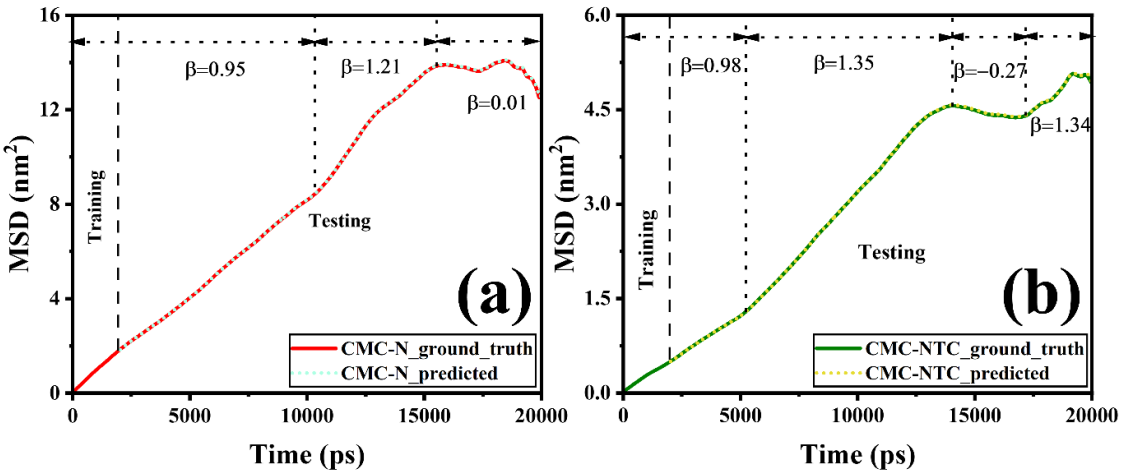


Figure S18. The MSD curve of oxygen atoms carboxyl functional gropup in the polymer backbone. (a) illustrate the MSD curve of CMC, (b) illustrate the MSD curve of CMC–N, (c) illustrate the MSD curve of CMC–NT polymer chain, and (d) represents the MSD curve of CMC–NTC polymer chain, I. schematic diagram of CMC, II. schematic diagram of CMC-N, III. schematic diagram of CMC–NT, and IV. schematic diagram of CMC–NTC



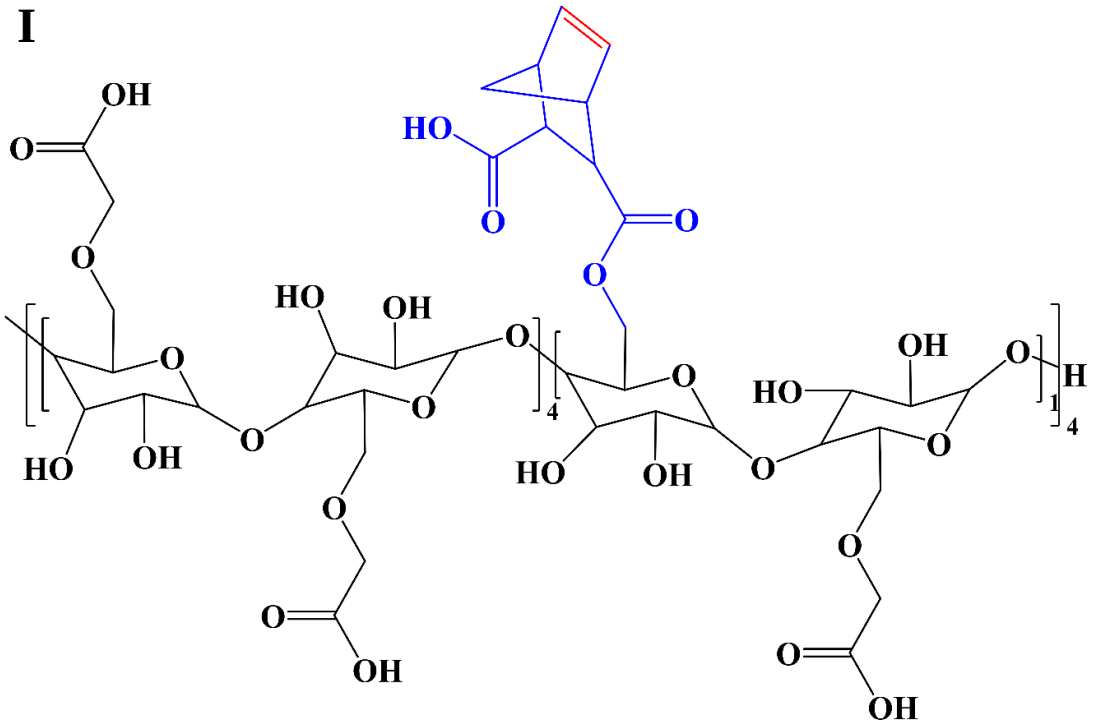
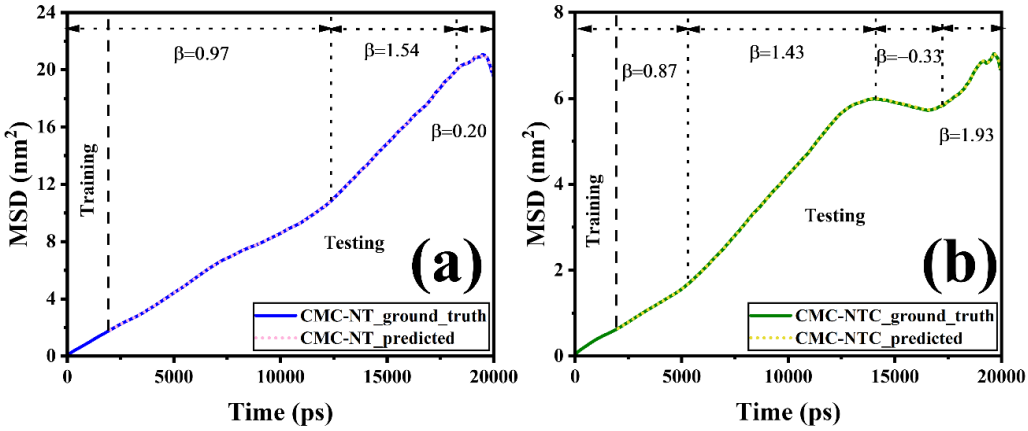


Figure S19. The MSD curve of unsaturated hydrocarbons presents in norbornene moieties in the polymer backbone. (a) illustrate the MSD curve of CMC–N, and (b) represents the MSD curve of CMC–NTC polymer chain, I. Schematic diagram of CMC– N.



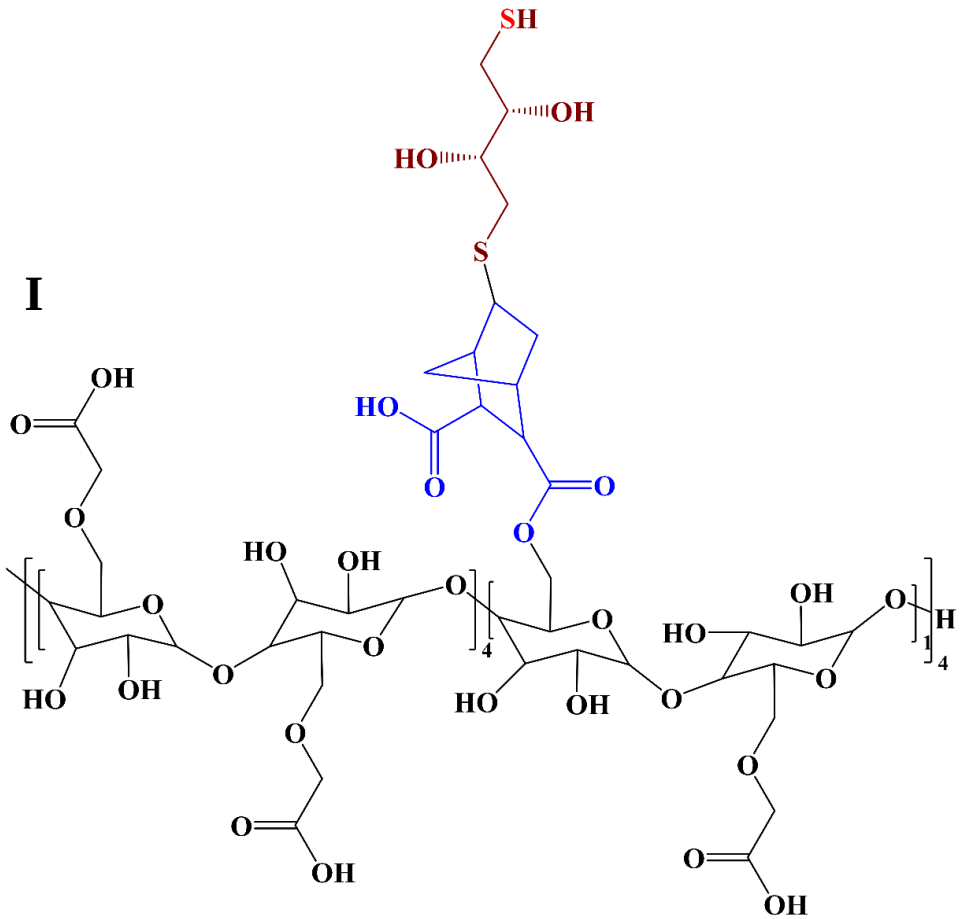


Figure S20. The MSD curve of unreacted thiol moieties presents in the polymer. (a) illustrate the MSD curve of CMC–NT, and (b) represents the MSD curve of CMC–NTC polymer chain, I. Schematic diagram of CMC– NT.

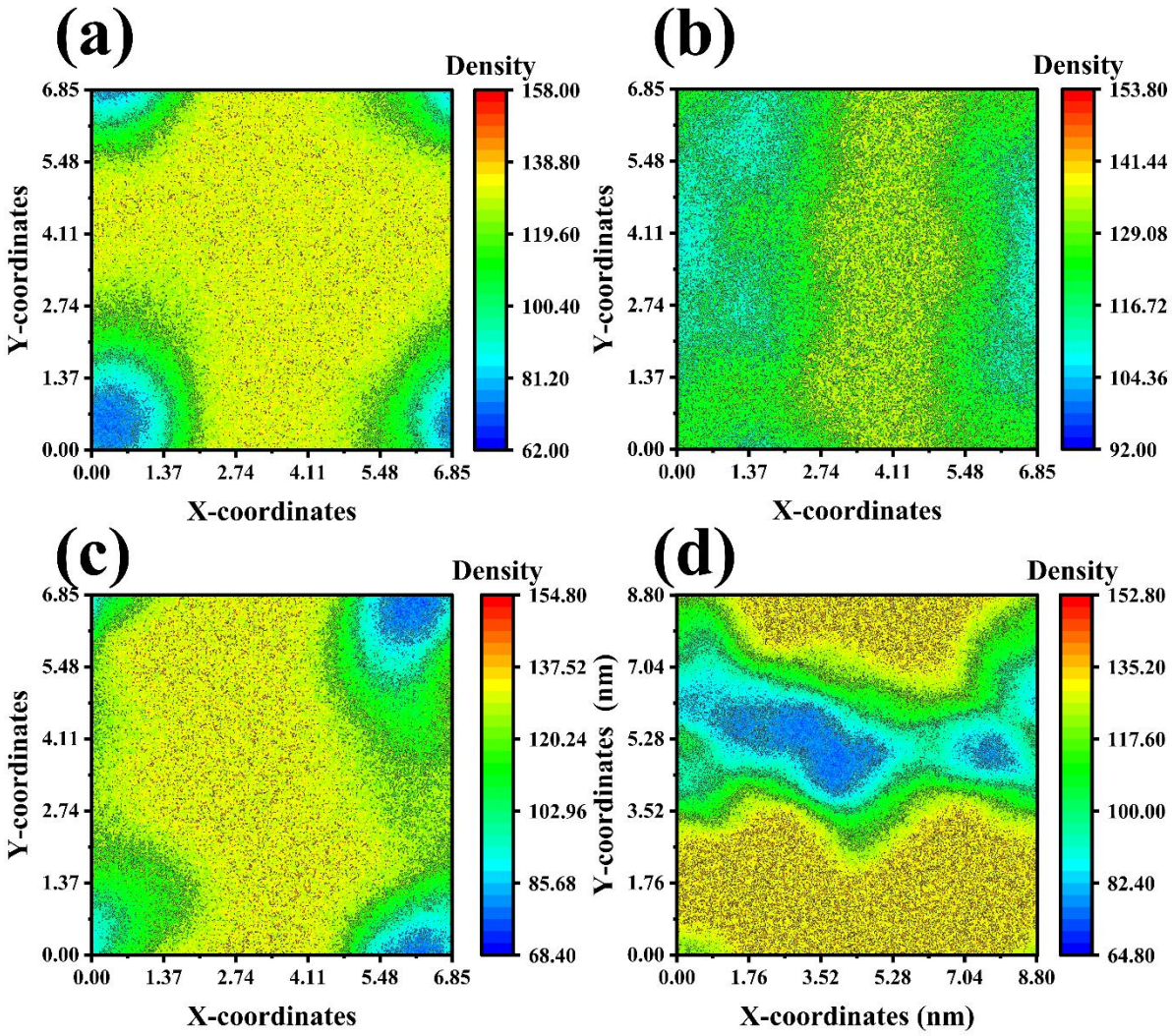
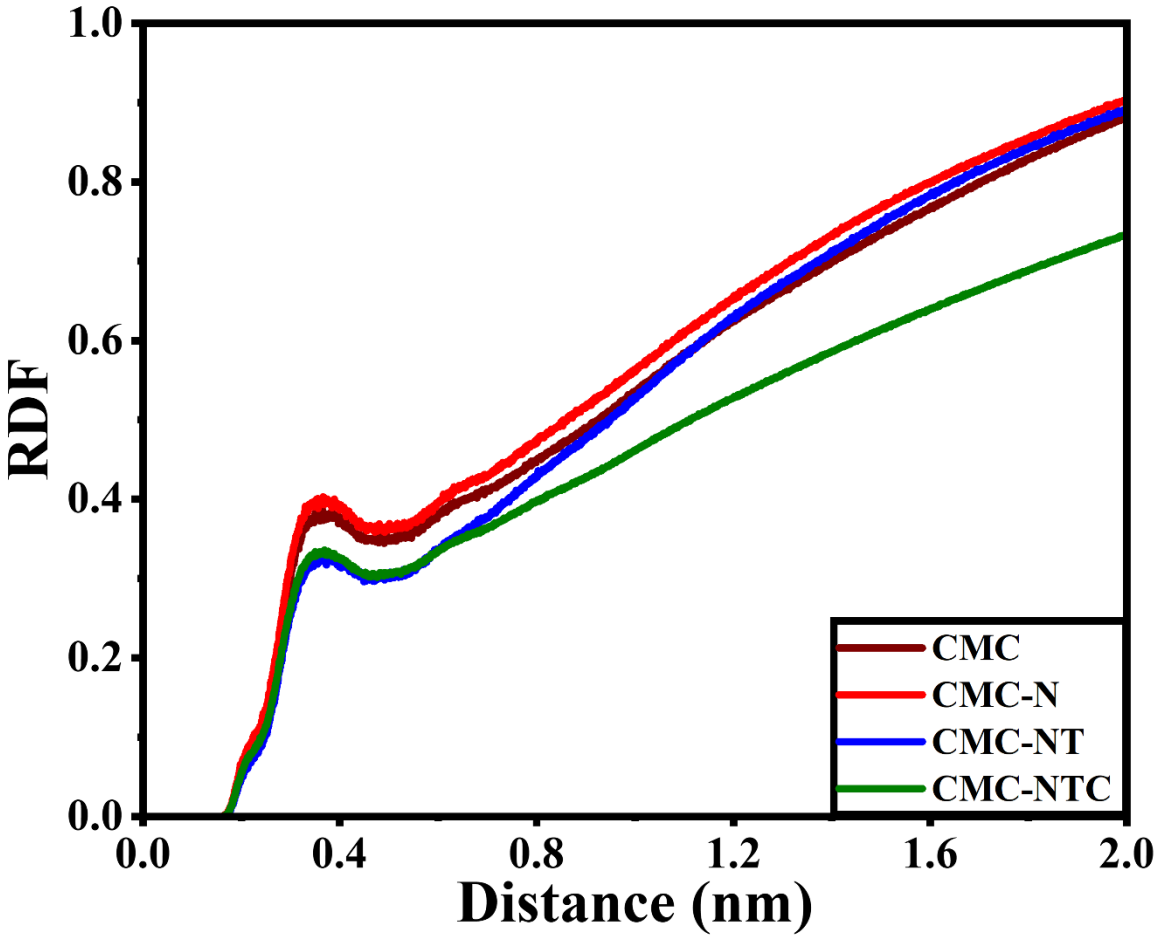


Figure S21. 2D number density for water contour plot (a) CMC (b) CMC-N (c) CMC-NT (d) CMC-NTC


RDF
Distance (nm)
0.0
0.2
0.4
0.6
0.8
1.0
0.0
0.4
0.8
1.2
1.6
2.0
CMC
CMC-N
CMC-NT
CMC-NTC

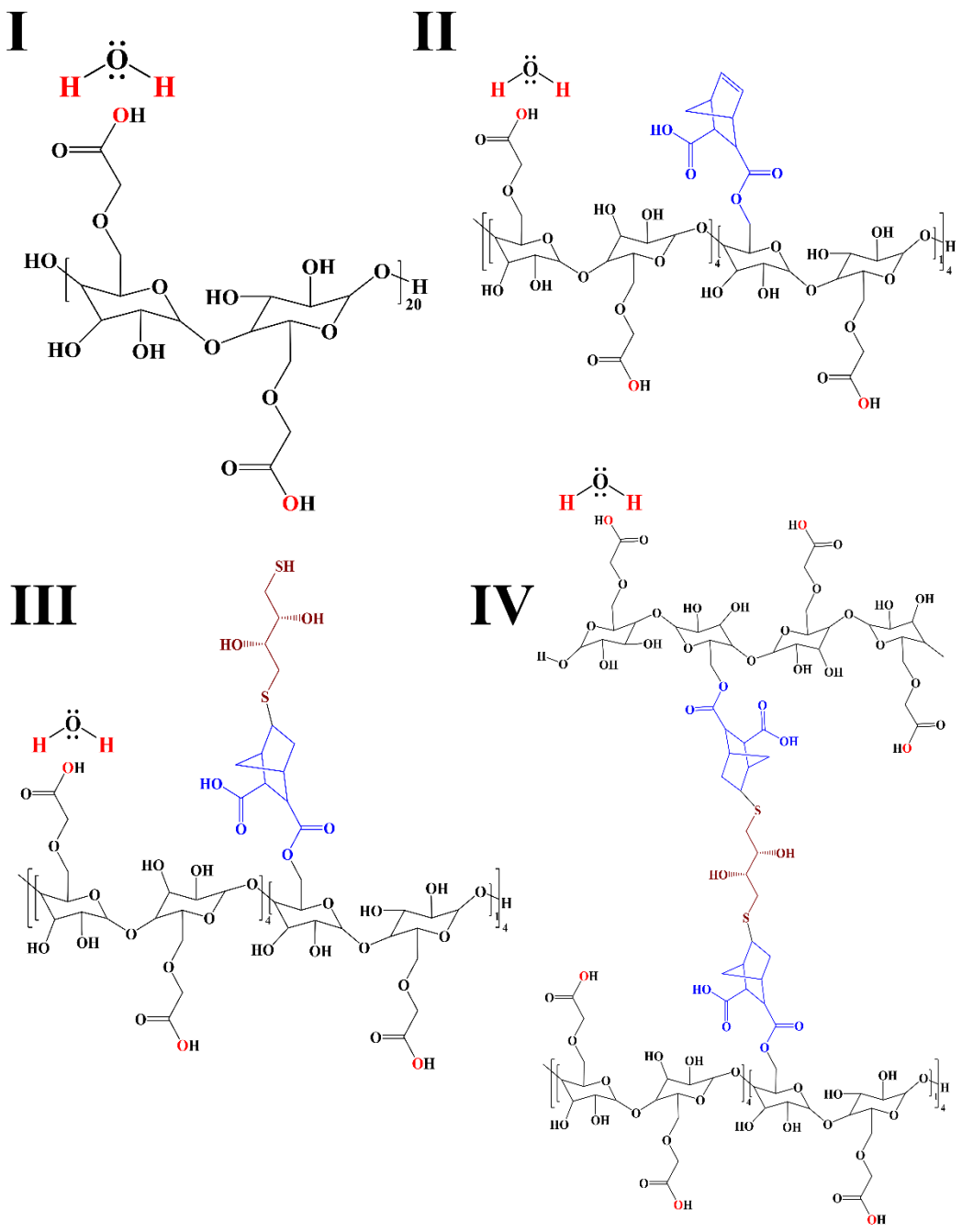
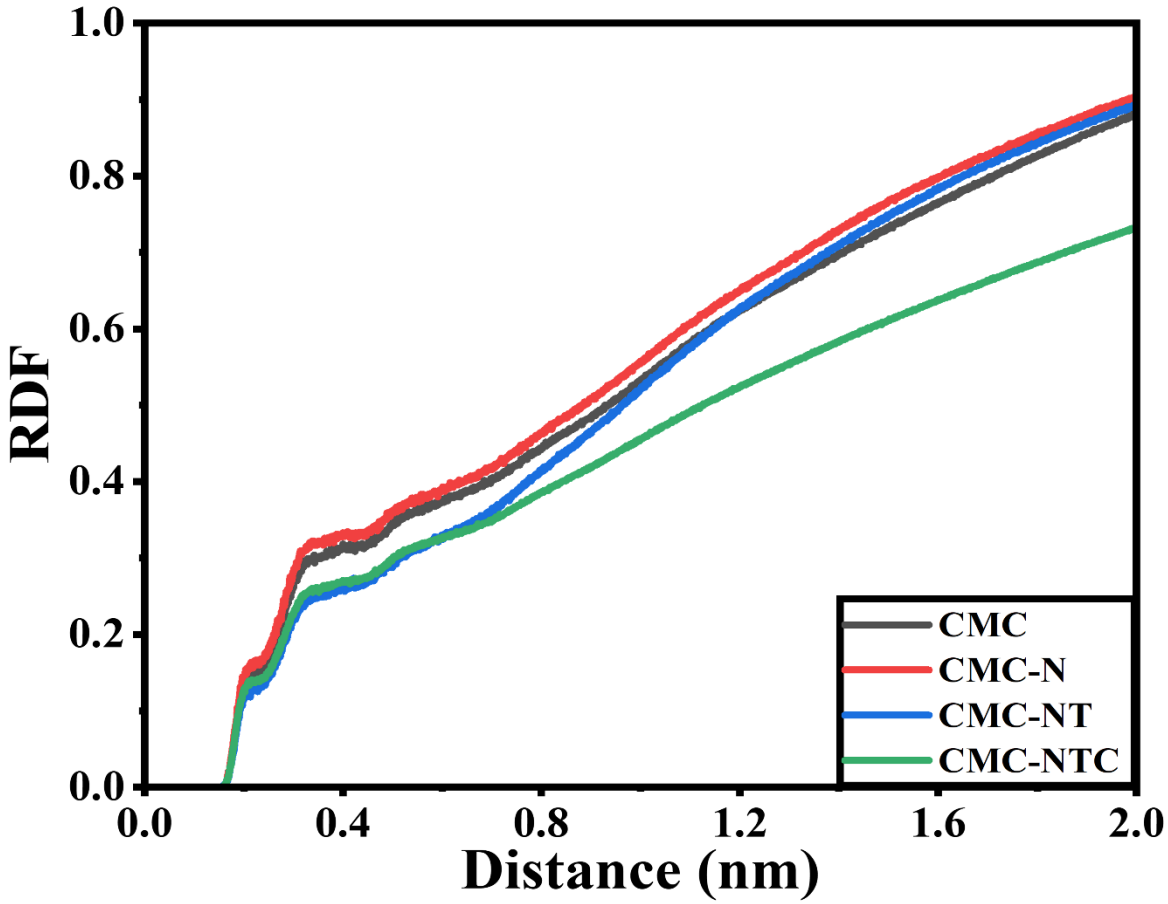


Figure S22. Radial distribution function of oxygen atoms in the carboxylic group of the polymer backbone to water hydrogen atoms. I. schematic representation of CMC interacting with water, II. CMC–N interacting with water, III. CMC–NT interacting with water, and IV. schematic representation of CMC–NTC interacting with water.


RDF
Distance (nm)
1.0
0.8
0.6
0.4
0.2
0.0
0.0
0.4
0.8
1.2
1.6
2.0
CMC
CMC-N
CMC-NT
CMC-NTC

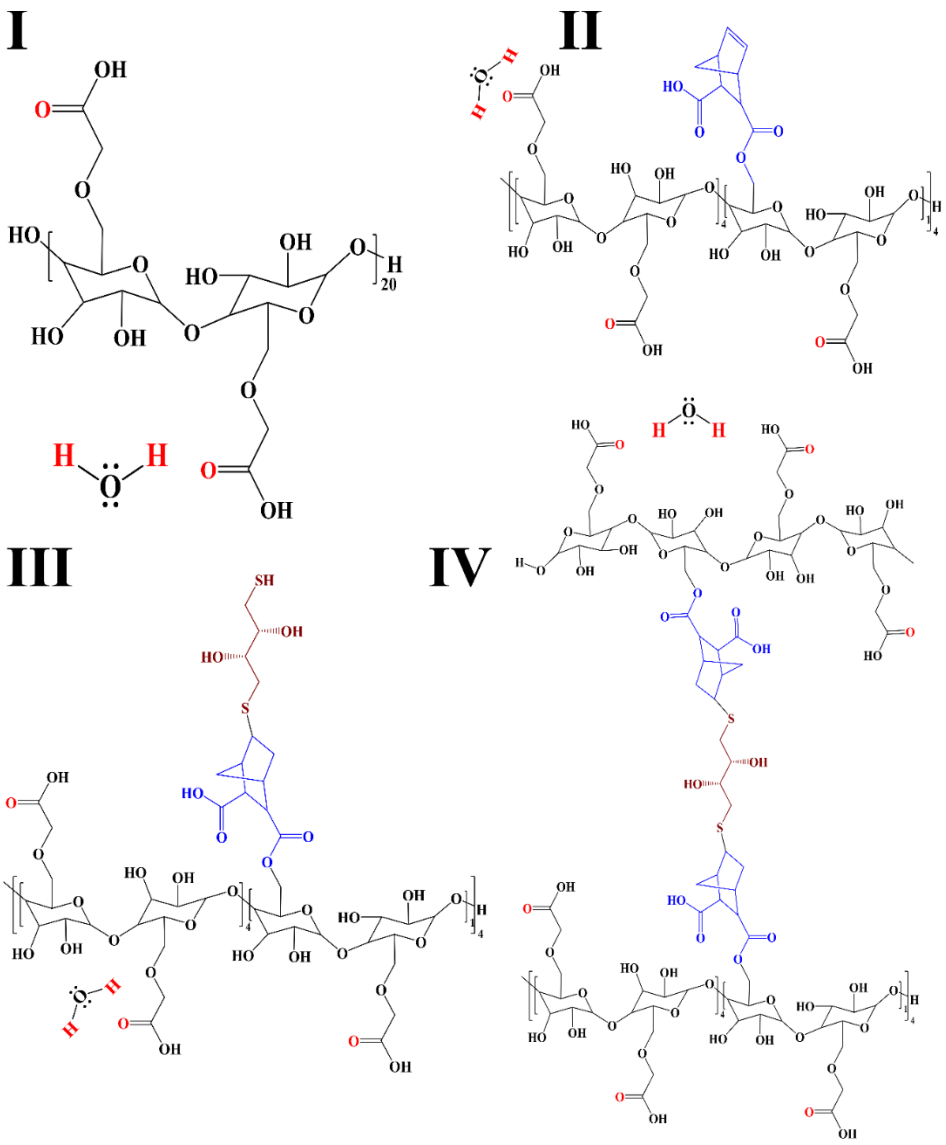
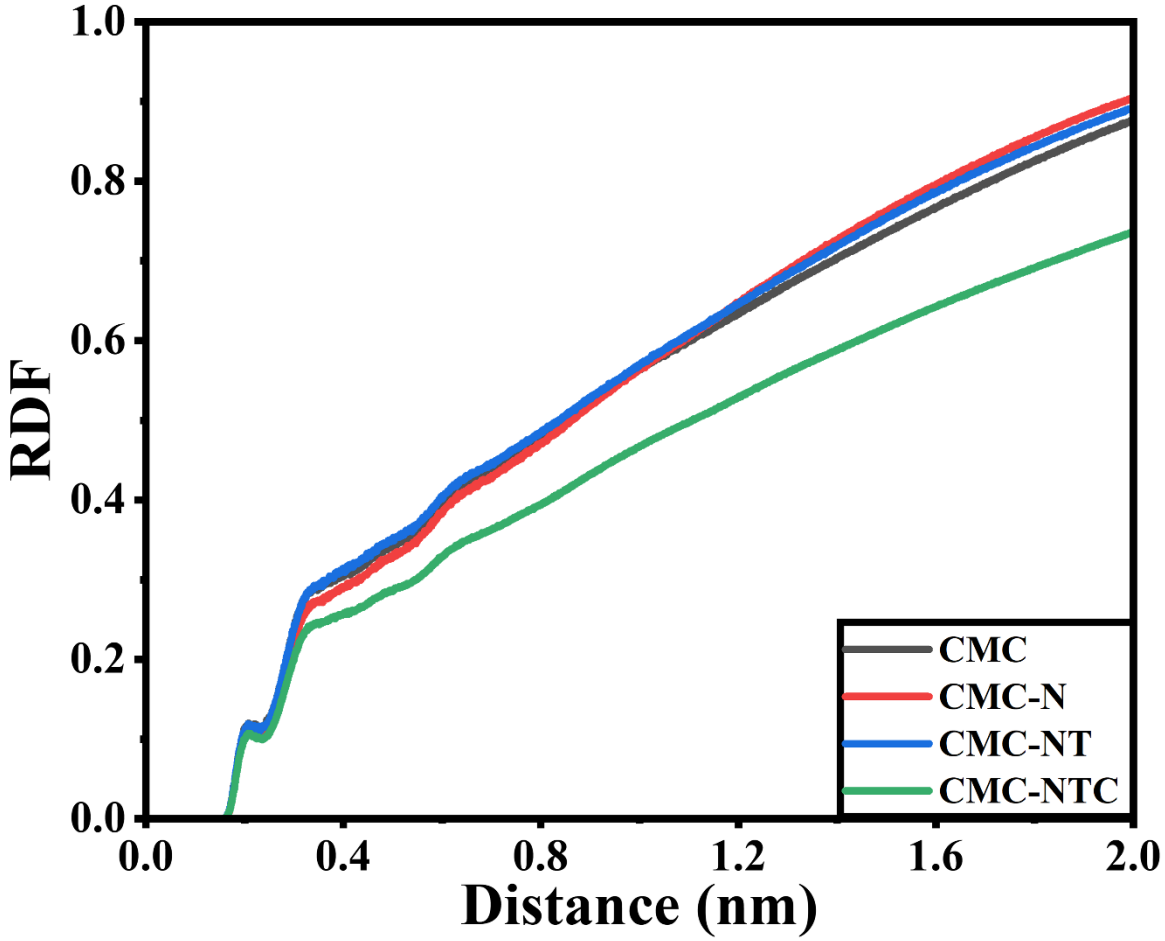


Figure S23. Radial distribution function of oxygen atoms of carbonyl group present in the polymer backbone to water hydrogen atoms. I. schematic representation of CMC interacting with water, II. schematic representation of CMC–N interacting with water, III. schematic representation of CMC–NT interacting with water, and IV. schematic representation of CMC–NTC interacting with water.


1.0
0.8
0.6
0.4
0.2
0.0
RDF
0.0
0.4
0.8
1.2
1.6
2.0
Distance (nm)
CMC
CMC-N
CMC-NT
CMC-NTC

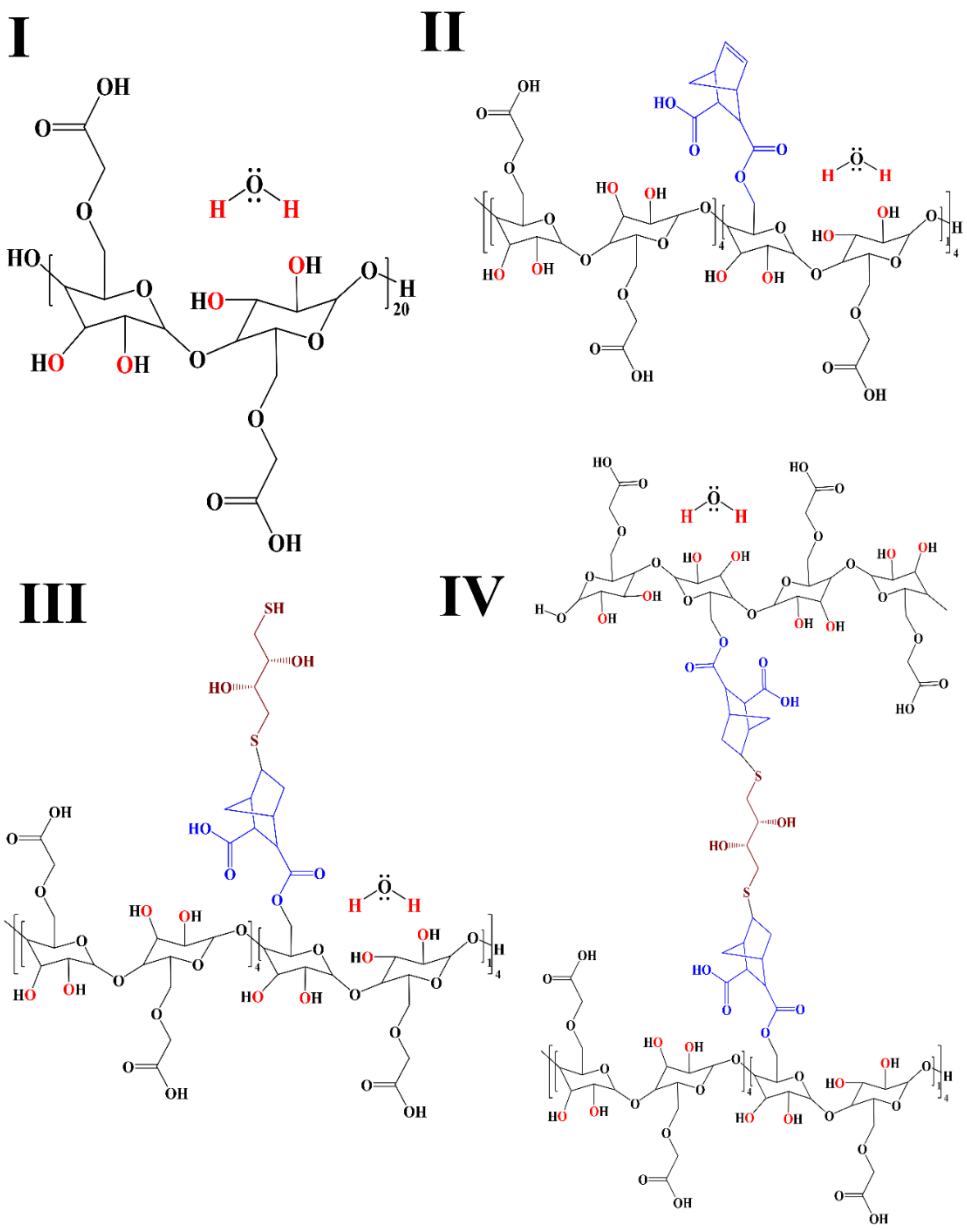
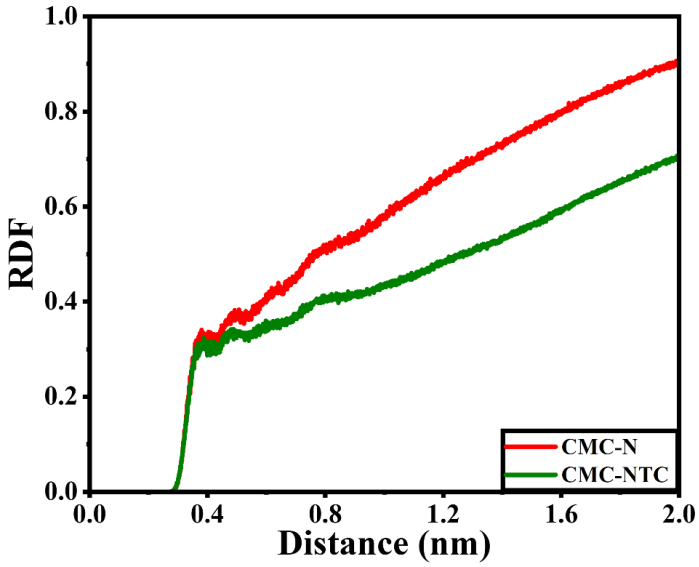


Figure S24. Radial distribution function of oxygen atoms of hydroxyl group present in the polymer backbone to water hydrogen atoms. I. schematic representation of CMC interacting with water, II. schematic representation of CMC–N interacting with water, III. schematic representation of CMC–NT interacting with water, and IV. schematic representation of CMC–NTC interacting with water.



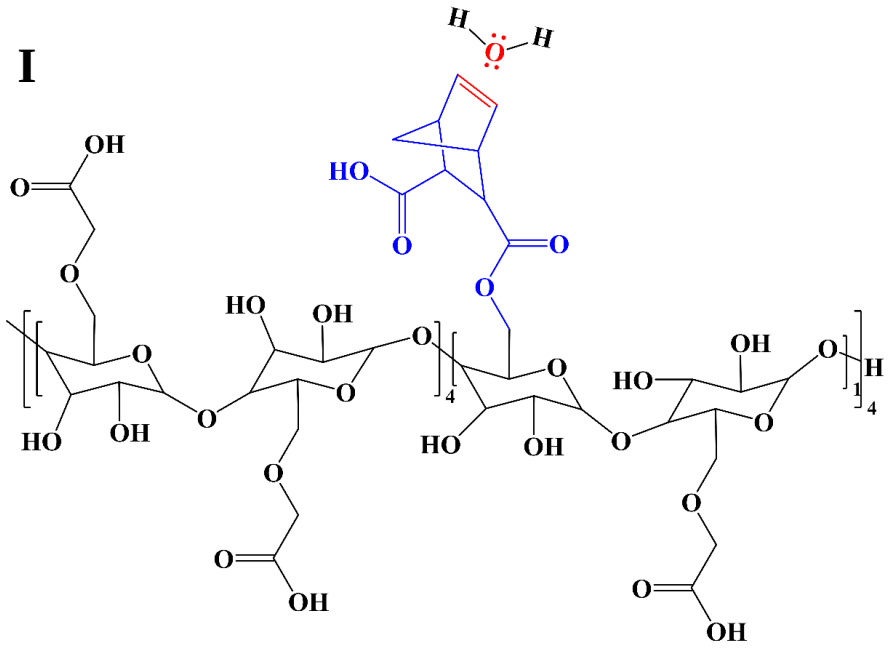


Figure S25. Radial distribution function of unsaturated hydrocarbons presents in the norbornene to water oxygen atoms (I) schematic representation of CMC–N interacting with water.

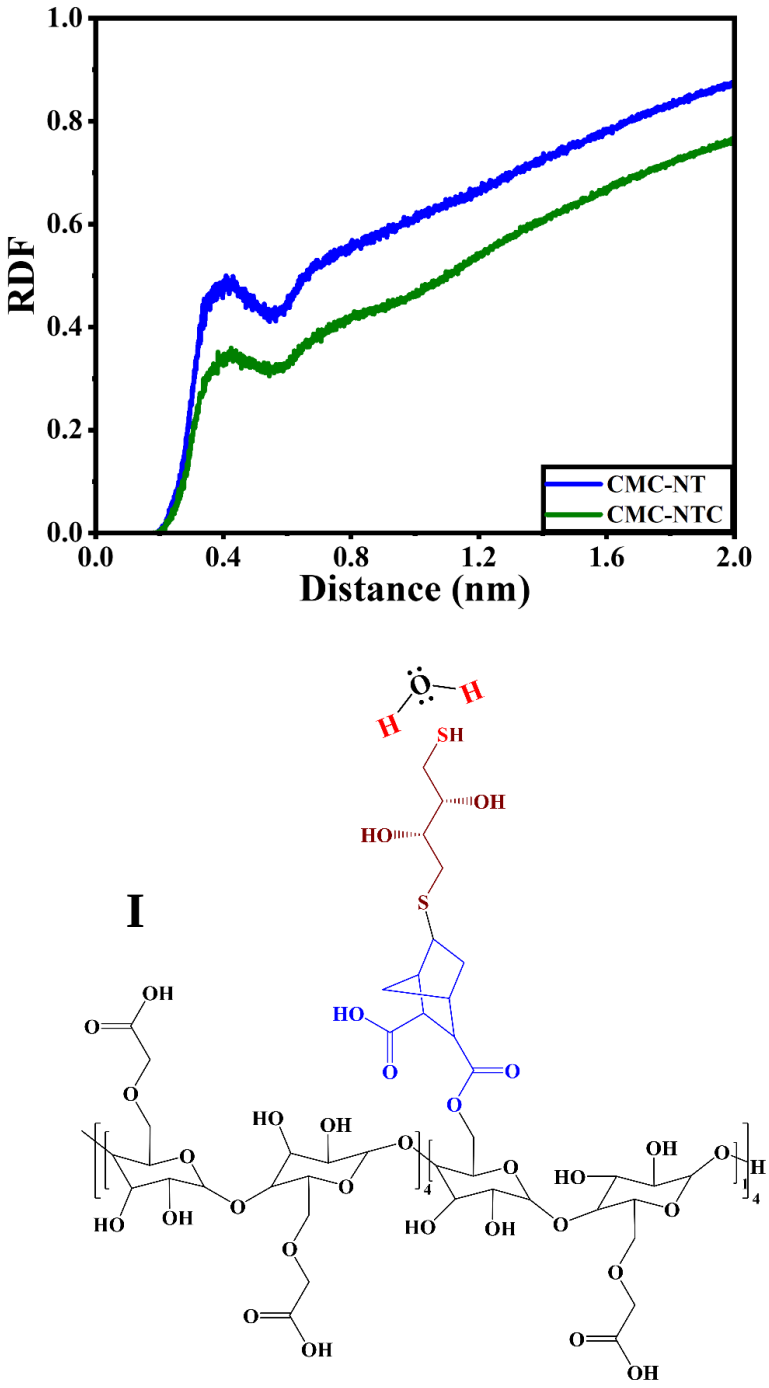


Figure S26. Radial distribution function of unreacted thiol moieties presents in the CMC–NT, and CMC–NTC to water hydrogen atoms and Schematic representation of thiol moieties in the CMC–NT interacting with water.